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Palaeomagnetism, rock magnetism and magnetic fabrics in Precambrian metamorphic terranes of the Huabei Shield, China

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Abstract

This study has investigated magnetic remanence, rock magnetism and anisotropy of magnetic susceptibility (AMS) in granulite and amphibolite grade metamorphic terranes of the Huabei Shield between Inner Mongolia in the west and the Bohai Sea in the east. Rock magnetic studies identify annealed metamorphic magnetite grains with multidomain properties as the remanence carriers; a widely recorded stable remanence was probably fixed by grain shape effects. Granulite facies terranes are typically between one and two orders more strongly magnetised than amphibolite terranes and AMS fabrics correlate mostly with metamorphic mineral fabrics observed in the country rocks. Progressive thermal demagnetisation identifies a range of two and three component structures resident in magnetite. An important component recognised as a partial or complete remagnetisation by Late Mesozoic–Tertiary tectonic/magmatic activity is present in basement at the southern margin of the outcrop (Miyun terrane) and where extensive granite plutonism has occurred (Zhunhua terrane). These components have directions corresponding to remanence in the Yunmeng Shan Granite (119–114 Ma, D/I = 33/58°, 39 samples, a_{95} = 3.5°, palaeopole at 201°E, 64°N). Most remanence elsewhere was probably acquired during post-tectonic uplift and cooling of the basement between ~2200 and 1850 Ma because palaeomagnetic directions are removed from the Phanerozoic palaeofield path and they are distinct from the palaeomagnetic record in the overlying Jixian Supergroup deposited at ~1840–900 Ma. These latter magnetisations are considered reliable indicators of the palaeofield during Late Palaeoproterozoic times because deformation of overlying supracrustal rocks is mostly slight and no prominent deflection of magnetic remanence by magnetic fabrics is observed. Palaeofield directions and poles attributed to the time of uplift-related cooling are: Qian'an Terrane (D/I = 215/71°, a_{95} = 9°, 17 samples, pole at 99°E, 10°N) and North Qianxi Terrane (D/I = 44/−45°, a_{95} = 4°, 41 samples, pole at 79°E, 11°S). In addition, a more widely-preserved shallow northerly component correlates with a NW → E swathe of components recorded by uplift-related cooling within the Datong–Huan'an granulite terrane in the west of the shield. A preliminary Palaeo-Mesoproterozoic apparent polar wander path for the Huabei Shield is defined from the Palaeoproterozoic record in the metamorphic basement rocks and the Meso-Neoproterozoic record in the overlying Jixian Supergroup. It incorporates a loop between ~2200 and 1850 Ma and exhibits a general east to west trend in subsequent times. © 1999 Elsevier Science Ltd. All rights reserved.

1. Introduction

Although the kinematics of the major Shields during Precambrian times is quantified by palaeomagnetic study, the recoverable record is always incomplete, both because the rock record is itself only sporadically preserved, and because later overprinting can obscure an ancient remanence record. The present Precambrian palaeomagnetic database is mostly derived from study of intense, but generally brief, igneous episodes and platformal successions of Meso- and Neoproterozoic age. The oldest palaeomagnetic record in each shield is preserved in the underlying basement rocks. This basement typically comprises a polymetamorphic gneiss complex produced by crustal accretion and repeated reworking during Archaean and Early Proterozoic times.

Whilst metamorphic rocks usually incorporate stable recorders of ancient remanence and hence provide scope for extending the palaeomagnetic record of the

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shields backwards in time, they have been little studied. The perceived problems include (i) the long time periods represented by uplift-related cooling, (ii) the difficulty of dating the magnetisations in the absence of available field tests and (iii) the presence of metamorphic fabrics which may bias the palaeomagnetic record in a way which cannot be readily quantified.

Initial magnetic remanence acquisition in Palaeoproterozoic metamorphic terranes is generally linked to the time interval of uplift-related cooling; this follows the last tectono-thermal episode responsible for reworking the basement and predates deposition of supracrustal successions deposited onto it (e.g. Morgan, 1976; Piper, 1985, 1992a). In some examples magnetisation can be specifically related to magnetite deposition in microfractures formed during decompression (Hay et al., 1988; Piper, 1992b). The remanence is then most obviously linked to age determinations dating isotopic closure in certain metamorphic minerals. The difficulties with more precise dating are offset by the ability of these terranes to record swathes of magnetisations embracing long time intervals during cooling (Morgan, 1976); they may then help to constrain whole segments of Proterozoic apparent polar wander (APW).

Fabrics in metamorphic rocks can be assessed by measurement of the anisotropy of magnetic susceptibility (AMS) in the palaeomagnetic cores. This is a bulk parameter recording preferred mineral orientation by metamorphic crystallisation and/or flow mechanisms. The grains contributing to this effect have paramagnetic and ferromagnetic properties; the paramagnetic minerals are mostly the iron-bearing silicates whilst the ferromagnetic contribution is from magnetite, and sometimes pyrrhotite or hematite. The strong ferromagnetic effect of the small volume of these latter grains is often much larger than the paramagnetic and (weak negative) diamagnetic effects of the bulk silicate matrix. Whilst the influence of fabric on the dilute amounts of ferromagnetic grains in which the stable remanence resides cannot be isolated by study of AMS, it is a useful qualitative indicator of bias of magnetic remanence by fabric; a comparison between results from samples of strong and weak fabrics at the same site is often able to recognise the likely importance of any such bias.

Medium to high grade polymetamorphic terranes occur along the northern margin of the Huabei (North China) Shield as inliers exposed beneath supracrustal successions comprising the Meso-Neoproterozoic Jixian Supergroup. Cooling of these terranes and their elevation to present surface levels occurred prior to deposition of the oldest part of the Jixian succession at ~1850 Ma. The remanence must however, have been acquired after the last stages of ductile deformation which occurred within the interval 2500–2200 Ma because regional temperatures were then above the Curie point. Currently the palaeomagnetism of formations within the Jixian Supergroup provides the main evidence for Precambrian APW of the Huabei Shield (Zhang (H.M.) et al., 1991, 1993). In addition palaeomagnetism and magnetic fabrics in the Datong–Huai'an granulite region of Inner Mongolia have been studied by Zhang (J.S.) and Piper (1994) and have made a preliminary contribution towards defining a pre-Jixian APW record. This paper reports results from terranes extending across the northern part of the shield towards the limit of major exposure north east of the Bohai Sea (Fig. 1). The collective results identify aspects of magnetic properties in this Precambrian basement and contribute towards extending the APW path of the Huabei shield backwards into the Palaeoproterozoic era.

2. Geological framework

The Huabei Shield is bounded to the north by the Mongolian (Central Asian) fold belt and to the south by the Qilian–Qinling–Dabie Shan. The Tancheng–Lujiang (Tanlu) fault runs through the eastern part and cuts the Dabie Shan belt. Although there has been much debate about the scale of movement on this fault boundary, no palaeomagnetically-detectable

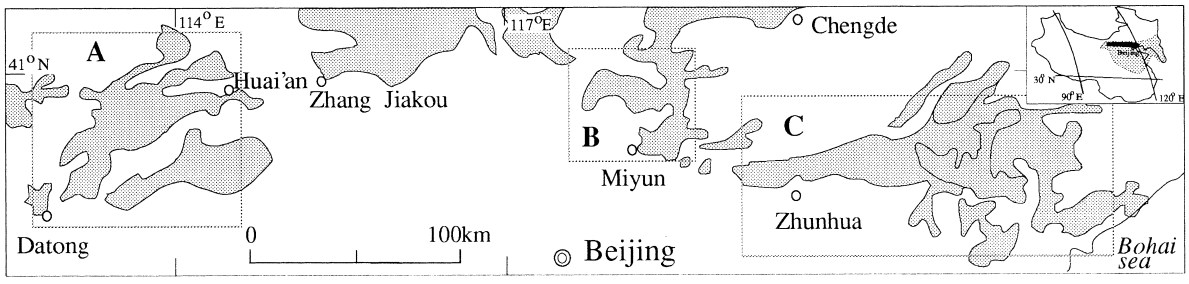


Fig. 1. Location map and regional outcrop (stippled) of Precambrian metamorphic basement in the North China Shield; see inset for the regional extent. Area A is the Datong–Huai'an granulite terrane from which palaeomagnetic results are described by Zhang (J.S.) and Piper, 1994). Areas B and C are the Miyun–Huairou and Zhunhua–Qianxi regions shown in more detail in Figs. 2 and 3.

motions between the China and Korean segments are identified since Upper Palaeozoic times (Doh and Piper, 1994). In the Alashan/Hexi Corridor region linking the Huabei and Tarim Shields, the oldest exposed rocks belong to the Longshoushan Formation thought to predate the Luliangian Orogeny at ~1850 Ma; these are intruded by 1790 Ma granites (Yang et al., 1986). As in other parts of the Huabei Shield, little evidence of volcanism or other signs of instability are observed in subsequent times suggesting that the cratonic elements were welded together to form a unified basement during the Luliangian Orogeny (Ma et al., 1980).

Geological evidence for Phanerozoic accretion of the North China Block (incorporating the Huabei Shield) with Siberia is still controversial and has been assigned to Middle Palaeozoic (Ren et al., 1987) and Mesozoic (Klimetz, 1983) times. Palaeomagnetic data suggest a location along the eastern rim of Gondwana during Late Precambrian times (Piper and Zhang, 1997). It had separated from Gondwana by Carboniferous times and palaeobotanic data imply close proximity to Eurasia by Upper Carboniferous times (Laveine et al., 1989). It subsequently moved as a composite block with Mongolia to become sutured to Siberia in Mesozoic times (Enkin et al., 1992). Convergence with the Eastern Asian assembly appears to have taken the form of anticlockwise rotation about an initial point of collision near the western end of the South Siberian Fold Belt (Zhao et al., 1996). Granites generated during late stages of suturing into the eastern Asian continental collage were emplaced as plutons into the metamorphic basement during Cretaceous times; they constitute the last major thermal event to influence this region.

Metamorphic basement of the Huabei Shield is exposed by uplift of a broad south-facing monocline and comprises outcrop in four major inliers, namely the Datong–Huai'an (Fig. 1A), Zhangjiakou–Xuanhua, Miyun–Huairou (Fig. 1B) and Zhunhua–Qianxi regions (Fig. 1C). The present paper is concerned mainly with the latter three inliers. Within a ca. 1000×200 km area extending from southern Inner Mongolia in the west to Hebei Province in the east (Fig. 1) basement outcrop averages some 75–85% gneisses and 15–25% supracrustal sediments (Qian et al., 1985). Metamorphism occurred during Archaean and Early Proterozoic times and the complex deformational histories are of comparable antiquity. North of the monocline dips of the Mid-late Proterozoic cover are mostly less than 10–15° and deformation since uplift has evidently been slight. Block faulting occurred during Late Mesozoic and Cenozoic times to develop fractures with a predominant NE–SW orientation; their locations are partly constrained to ductile shear belts of Lower Proterozoic age but brittle effects are concentrated within a few tens up to a few hundreds of metres of the fault zones.

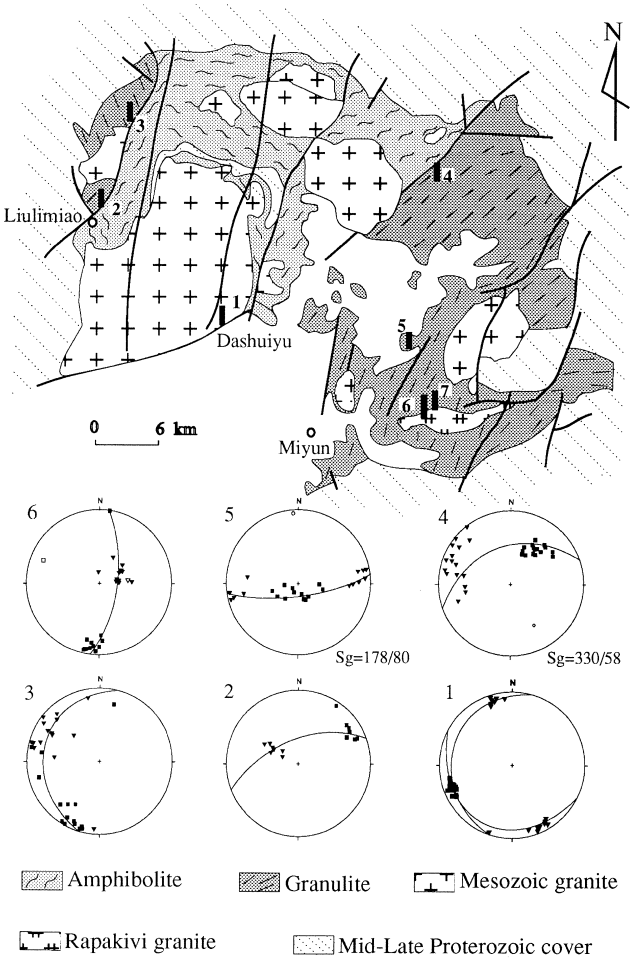


Fig. 2. Lithological and sampling site map of the Miyun metamorphic terrane. The numbered small black columns are the locations of the palaeomagnetic sites. The projections show AMS fabrics defined by directions of maximum (black squares) and intermediate (black triangles) susceptibility plotted as lower hemisphere projections. The small open symbols are poles to the gneissose foliation and the great circles are planes of magnetic foliation defined by k_1 and k_2. Site 6 is in a Mesoproterozoic rapakivi intrusion.

The Miyun–Huairou terrane (Fig. 2) comprises predominant amphibolite facies extensively intruded by Mesozoic granites; granulite facies rocks are upfaulted at the east and west sides. Zircon U–Th–Pb and U–Pb age determinations on the metamorphic rocks are in the range 2567–2346 Ma (Zhou et al., 1993). Petrologic studies show that metamorphic conditions attained temperatures of 650–700°C in the amphibolite facies and 800 ± 30°C at pressures of 8–13.5 kb in the granulite facies (Lu, 1991).

High grade rocks have a more widespread distribution in the Zhunhua–Qianxi terrane (Fig. 3) where isotopic data define basement uplift and deposition of the first cover rocks at ~1900 Ma (Sun, 1984). The

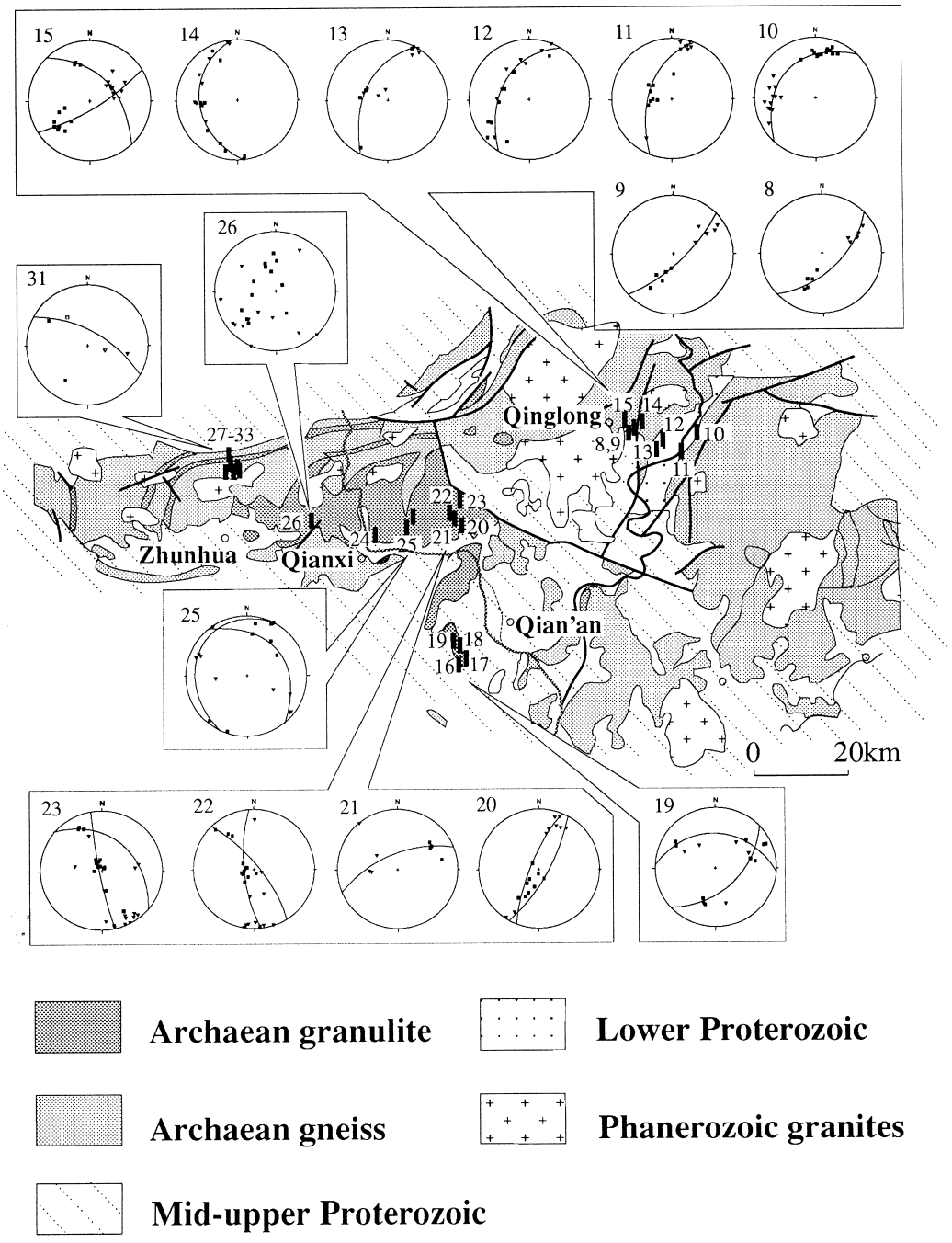


Fig. 3. Lithological and sampling site map of the Zhunhua–Qianxi metamorphic terrane (see Fig. 1) with projections of site AMS fabrics. Symbols are as for Fig. 2.

basement rocks are subdivided here into three groups comprising:

(i) Granulite facies (the Qianxi Group) locally retrogressed to amphibolite facies. The major rock type is intermediate-acid gneiss derived from a protolith of tonalite–trondjemite–granodiorite (TTG) affinity. Mafic to ultramafic granulites occur as inclusions; sillimanite–garnet gneisses (khondalites) and some interbedded supracrustal sediments of banded iron-stone formation (BIF) are also present. Temperatures of granulite facies metamorphism reached 739–1068°C (Zhang (R.Y.) and Cong, 1981; Wang (R.M.) and Chen, 1982; Wang (R.M.) et al., 1985; Bai and Yang, 1984; Qian et al., 1985) at pressures of 11.9–14.2 kb (Zhang (R.Y.) and Cong, 1981; Qian et al., 1985) or 8–11 kb (Bai and Yang, 1984) (Fig. 4).

(ii) An amphibolite facies association, composed mainly of biotite–plagioclase gneisses distributed

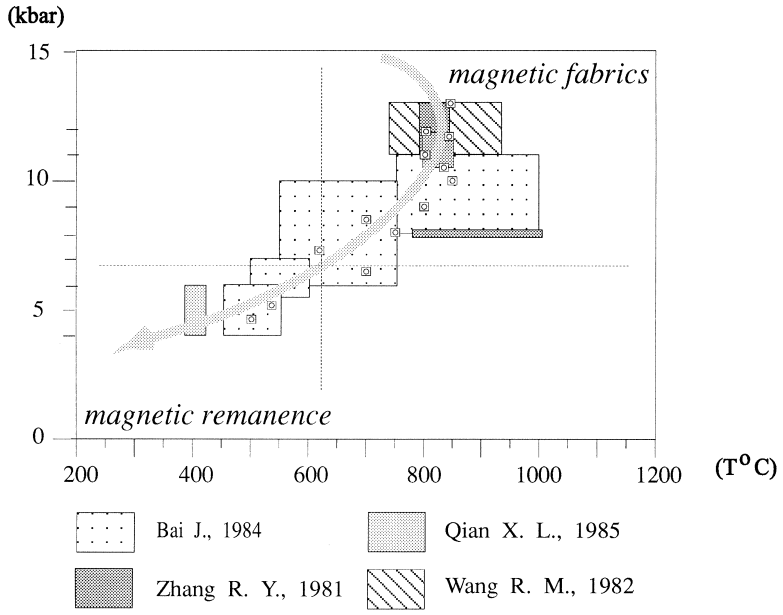


Fig. 4. The temporal change in P–T conditions in the Zhunhua–Qianxi area (see Fig. 1 (C), 3) compiled from the results of senior authors as indicated (Bai and Yang, 1984; Qian et al., 1985; Zhang and Cong, 1981; Wang and Chen, 1982). Also shown are approximate stages in the metamorphic evolution of the basement to which the magnetic fabric and magnetic remanence acquisition are assigned.

around a high grade core. The metamorphic facies show transition outwards from high amphibolite to epidote amphibolite grades and record peak conditions of 500–760°C and 6–10 kb (Fig. 4).

(iii) A volcanic-sedimentary succession, metamorphosed to lower amphibolite–epidote facies and distributed along a narrow NNE trending zone in eastern Qinglong County; an unconformity at the base is defined by a metamorphosed conglomerate layer. This succession is a component of an early Proterozoic orogenic belt at the margin of the craton formed at ~2500 Ma but deformed and metamorphosed at ~2000–2100 Ma when metamorphic conditions (see Fig. 4) attained 500–600°C at 5.5–7.0 kb in lower amphibolite facies and ~500°C at 4–6 kb in the epidote-amphibolite facies (Bai and Yang, 1984).

The igneous protolith at the core of these terranes is probably of Early-mid Archaean age as defined by age determinations between 3470–2980 Ma (Jahn et al., 1987, Liu et al., 1990, 1992). TTG facies within grey gneisses of the Datong–Huai'an terrane in the west are of comparable antiquity whilst the Qianxi Group has yielded 3850–3300 Ma ages (U–Pb zircon results of Liu et al. (1985, 1992), and Sm–Nd isochron determinations of Jahn and Zhang (1984), Jahn and Ernst (1990) and Liu et al. (1990). Collectively these studies identify earliest deposition of supracrustal rocks at 3500–3300 Ma following a crust-forming event.

Age determinations of the granulite facies metamorphism are predominantly in the range 2460–2517 Ma (Rb–Sr and Sm–Nd isochron ages, Bai and Yang, 1984; Cui et al., 1989, Jahn et al., 1987; Liu et al., 1985). Ages derived from the amphibolite facies are comparable to those from the granulites and in the range 2450–2650 Ma (Bai and Yang, 1984; Cui et al., 1989) suggesting that the higher grade amphibolite facies metamorphism rapidly succeeded granulite facies in Late Archaean–Early Proterozoic times. Metamorphism at lower amphibolite to epidote grades is dated at 2080–2200 Ma (Bai and Yang., 1984) broadly contemporaneous with polyphase granitoid intrusion and pegmatite emplacement at around 2200 Ma (Liu et al., 1985; Sun, 1984).

The time sequence of metamorphic conditions, as summarised in Fig. 4 from data in the Miyun and Zhunhua–Qianxi regions, defines a clockwise P-T-t scheme very similar to that resolved from the Datong–Huai'an region to the west (Liu et al., 1992, Zhang (J.S.) and Piper, 1994; Zhang (J.S.) et al., 1994). Deformation and metamorphic events of the prograde phase have been largely obliterated by the peak events which produced tight N–S folds in the Miyun and Zhunhua–Qianxi regions; later deformation in the lower grade rocks to the east produced folds with a NNE trend. The P-T-t trajectory is comparable to other Precambrian gneiss complexes (Hartley, 1988) with the clockwise trajectory generally interpreted to reflect crustal thickening during a compressional phase, possibly in a collisional setting, followed by extensional collapse.

Temperatures dropped below the Curie points of the magnetic carriers (~580°C for magnetite) during

decompression and uplift following the 2100–2200 Ma events. The metamorphic terranes would subsequently have cooled to ambient temperatures by the time of deposition of basal members of the Jixian Supergroup (~1850 Ma) which are unmetamorphosed. Cooling of the metamorphic rocks through the Curie point therefore attributes any primary remanence to the time interval ~2200–1850 Ma.

3. Field and laboratory methods

Cores were drilled at 32 locations in the metamorphic rocks, plus igneous intrusions, predating deposition of the Jixian Supergroup as summarised in Table 1. In addition the largest member of the Mesozoic granite-granodiorite suite, the Yunmeng Shan Granite, was sampled at Site 1 to assess the importance of the Mesozoic thermo-tectonic event in overprinting the basement rocks. A sampling site comprises between seven and twenty two 2.4 cm diameter cores distributed across at least several metres of outcrop (e.g. Fig. 7). Cores were oriented in situ by sun and magnetic compasses and cut into 2.2 cm length cylinders for magnetic measurements.

Magnetic susceptibilities were initially measured using a Bartington bridge and AMS fabrics determined by "Minisep" anisotropy delineator. Cores were then measured by "Minispin" magnetometers before treatment and following each stage of progressive cleaning by thermal demagnetisation. The latter employed Magnetic Measurements demagnetisers in steps of 50 or 100°C to 500°C and subsequently in steps of 20°C to the Curie points of the magnetic carriers; the

Table 1
Summary of palaeomagnetic sites in metamorphic terranes of the North China Shield

Site No.	Rock type	J_o ($\times 10^{-5}$ A m^2/kg)	k_1/k_3	Shape
(i) *Liulimiao area*				
1	Yunmeng Shan Granite	0.44–9.86	1.30–1.48	P
2	Basic dyke	4.07–39.40	1.33–1.76	O
3	Granulites in granite aureole	0.57–8.25	1.02–1.07	P/O
(ii) *Miyun Region*				
4	Granulites	0.50–28.10	1.14–1.30	P/O
5	Basic lenses in granulite	4.42–77.78	1.12–1.28	O
6	Contact with Rapakivi Pluton	0.73–37.09	1.04–1.45	P
7	BIF and felsic granulite	376.90–1519.51		
(iii) *Qinglong–Qian'an Region*				
8	Lavas cut by dyke site 9	0.03–1.17	1.05–1.12	O
9	Basic dyke, amphibolitised	< 0.01	1.05–1.20	O
10	Amphibolitised Volcanics	0.34–2.62	1.03–1.51	O
11	Amphibolitised Volcanics	0.03–1.42	1.04–1.31	O
12	Amphibolitised volcanics	< 0.01–0.04	1.17–1.24	O
13	Basic sheet in volcanics	0.01–0.23	1.10–1.16	O
14	Amphibolitised volcanics	< 0.01–0.05	1.10–1.25	O
15	Amphibolitised volcanics	0.01–0.05	1.09–1.13	O
16	Dyke cutting TTG	1.02–47.50	1.03–1.04	O/P
17	TTG granulite	0.03–0.42	1.04–1.12	O
18	Banded Ironstone Formation	64.21–504.66		
19	Monzonitic granite	0.48–122.20	1.04–1.40	O/P
(iv) *Qianxi–Qian'an Region*				
20	Banded granulites	1.02–29.44	1.31–1.52	O
21	Granulite	6.90–12.45	1.01–1.19	O
22	Granulite	1.03–15.21	1.20–1.51	P
23	Massive felsic granulites	7.53–51.92	1.04–1.72	O/P
24	Granulites	1.55–61.93	1.05–1.88	O
25	Basic dykes	3.47–40.74	1.06–1.62	O
26	Granulites	0.13–158.40	1.01–1.68	O
(v) *Zhunhua Region*				
27	NE–SW basic dyke, cuts 30	17.79–1012.83	1.02	O
28	NE–SW thick basic dyke	32.89–109.93	1.02–1.04	O
29	Basic dyke, cuts 30	18.39–56.67	1.01–1.03	O
30	Acid intrusion, pre-Jixian	2.25–10.53	1.04–1.06	O
31	Granulite	1.98–32.89	1.01–1.15	O
32	Felsic Granulite	7.00–59.87	1.68–2.10	O
33	Basic Dyke, pre-Jixian	69.21–87.54	1.03–1.19	O

The dominant AMS ellipsoid form is indicated as O = Oblate and P = Prolate

Table 2
Petrological properties of typical metamorphic rocks and comparison with susceptibility and magnitude of AMS

Site No.	Rock Type:	Mineral (mafic)	Assemblage (felsic)	k (av.)	AMS (av.)
32	Felsic granulite	Gnt (15%)			
		Bi (5%)	F + Q (>70%)	1996	1.68
23	Felsic granulite	Bi (5–10%)			
		Opx (10%)	F + Q (70–80%)	2160	1.22
23	Basic granulite	Hb (60%)			
		Opx (10%)			
		Cpx (5%)			
		Mag (3%)	F (15–20%)	56384	1.56
17	Felsic granulite	Bi (5%)	Q + F (>90%)	355	1.13
17	Basic granulites	Hb (50%)			
		Cpx (30%)	F (15–20%)	820	1.09
5	Ultramafic Incl.	Cpx (35%)			
		Hb (45%)			
		Mag (8%)	F (<10%)	219630	1.21
6*	Rapakivi contact		Oligoclase (>85%)	40	1.15

* Sample from acid vein in granulite at contact. Gnt = garnet, Bi = biotite, Opx = orthopyroxene, Cpx = clinopyroxene, Hb = hornblende, Mag = magnetite/ulvospinal/ilmenite, F = feldspar and Q = quartz

ongoing procedure was conducted within a nullified field environment provided by a large set of Rubens coils to minimise secondary remanence acquisition. Components constituting the Natural Remanent Magnetisation (NRM) in each sample were resolved by study of orthogonal projections. Characteristic magnetisations (ChRM's) were isolated and their equivalent directions calculated by principal component analysis.

3.1. Magnetic properties

A general feature of the intensities of magnetisation and susceptibilities in Precambrian metamorphic terranes is a range of several orders of magnitude, controlled by variations in dilute amounts of ferromagnetic constituents (Tables 1 and 2). This is also expressed in contrasts between felsic and mafic granulites (Fig. 5a). It is less marked between acid and basic intrusives; predictably the banded ironstones are by far the most strongly magnetised facies. The ferromagnetic content depends on both the source and the metamorphic grade. Thus the most important regional contrast in North China is between the strongly magnetised granulite facies terranes and the weakly magnetised amphibolite facies rocks of the Qinglong terrane (Table 1 and Fig. 5b). Although metamorphic rocks in the latter region are derived mostly from a basic igneous source, the iron is present in paramagnetic amphiboles and chlorites, and magnetisations are typi-

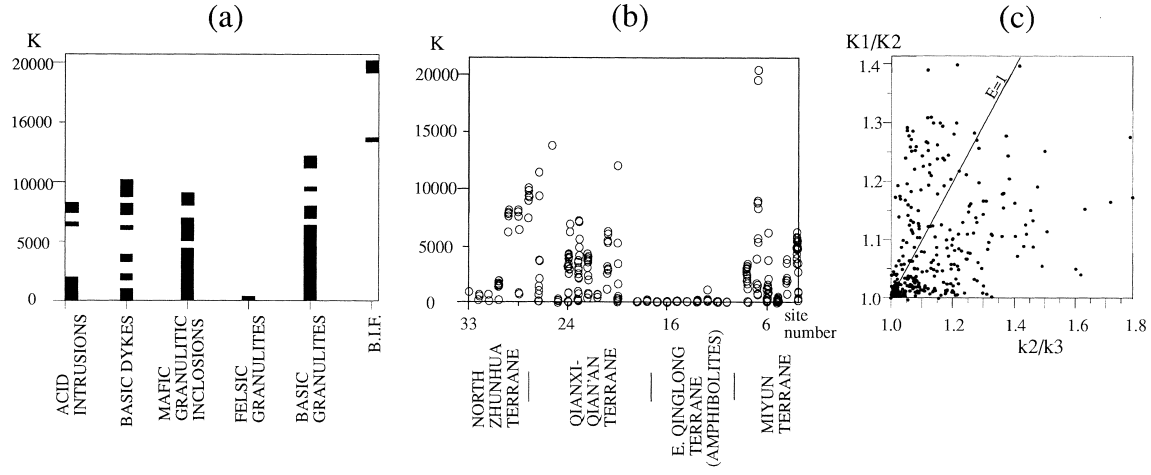


Fig. 5. (a) The distribution of magnetic susceptibility (SI units) in the metamorphic facies of this study. (b) Distribution of magnetic susceptibilites in the different sample areas; note the contrast between values in the amphibolite facies terrane (Sites 8–15 and the granulite facies elsewhere. (c) The magnitude of k_2/k_3 (magnetic foliation) against k_1/k_2 (magnetic lineation) in metamorphic rocks of this study. The line E = 1 separates oblate (foliated) fabrics on the right from prolate (lineated) fabrics on the left; note that oblate fabrics tend to predominate.

Table 3
Summary results of rock magnetic investigations

	Thermomagnetic:		KLT:			KHT:	Hysteresis:			
Site	T_c	Curve	Group	RS	ΔS	Curve	M_s	M_{rs}/M_s	X_{MD}	Type
Granulites										
4	585	2a	3a	0.39	0.97	H	226.2	0.031	97	M
6	590	2a	3a	0.36	0.97	H	185.4	0.019	100	M
22	570	2a	3b	1.04	0.12	H	279.3	0.015	100	M
24	585	2a	3b	0.51	0.89	H	91.2	0.141	75	M
26	590	2a	3b	1.43	0.83	H	39.1	0.094	85	M
31	580	2a	3b	1.09	0.79	H	344.3	0.007	100	M
Banded Ironstones										
7	595	3	3a	1.40	1.02	H	289.2	0.016	100	M
18	600	3	3a	1.18	0.98	H	265.6	0.010	100	M
26*	595	3	3a	1.32	1.02	H	344.4	0.027	99	M
Amphibolites										
9	590	2c	2	2.48	—	—				P
10	595	2c	2	3.15	—	—	28.6	0.057	—	P
12	590	2c	2	—	—	—	16.6	0.0	—	P
13	590	2c	2	2.74	—	—	14.7	0.058	—	P
14	595	2c	2	2.41	—	—	17.5	0.063	—	P
15	590	2c	2/3	1.36	0.11	—	9.2	0.001	—	P

cally 2–3 orders weaker than in the granulites. Precipitation of primary metamorphic magnetite appears to account for much stronger magnetisations in the granulite facies (cf. Tables 1 and 2).

Rock magnetic properties of granulite, amphibolite and BIF samples were studied to evaluate the nature of magnetic carriers and compare properties of the metamorphic grades; the results are summarised in

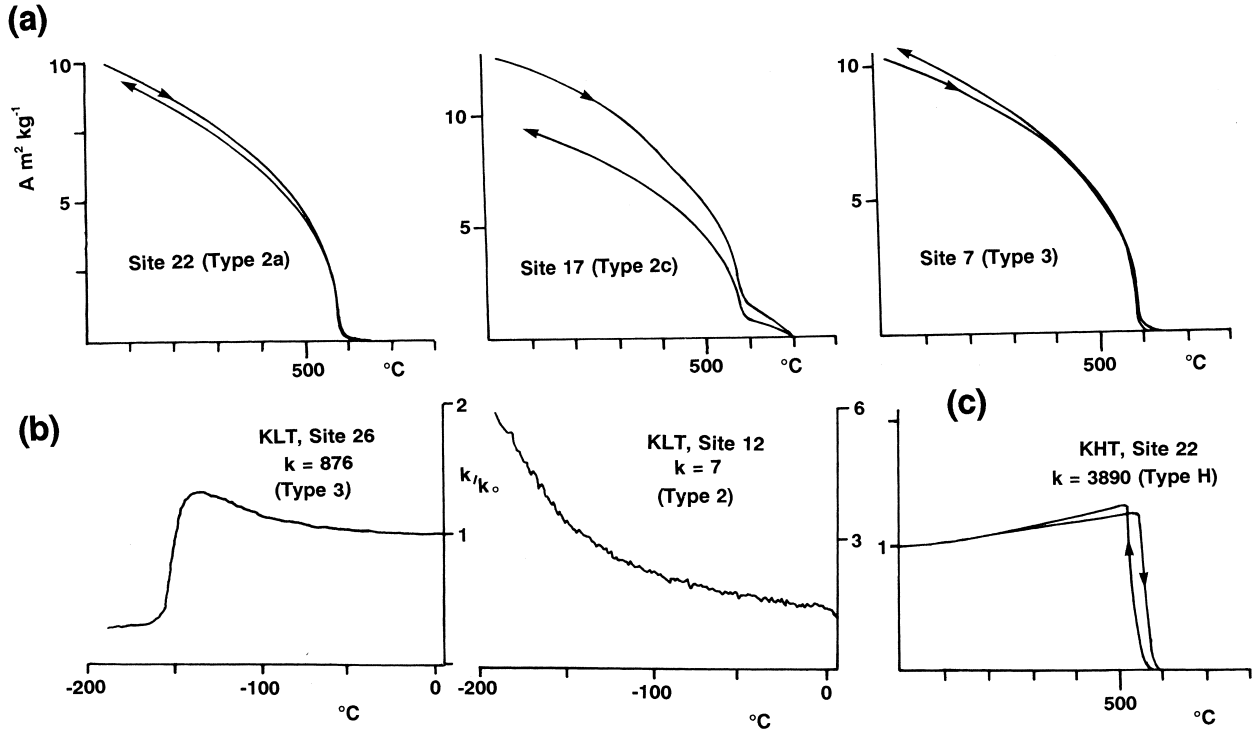


Fig. 6. (a) Typical thermomagnetic curves (saturation magnetisation, J_s, versus temperature T) derived from the granulite facies. (b) Typical KLT curves from granulite facies showing variation of susceptibility at low temperatures. (c) KHT curve from granulite showing variation in susceptibility at high temperatures. See text for explanation.

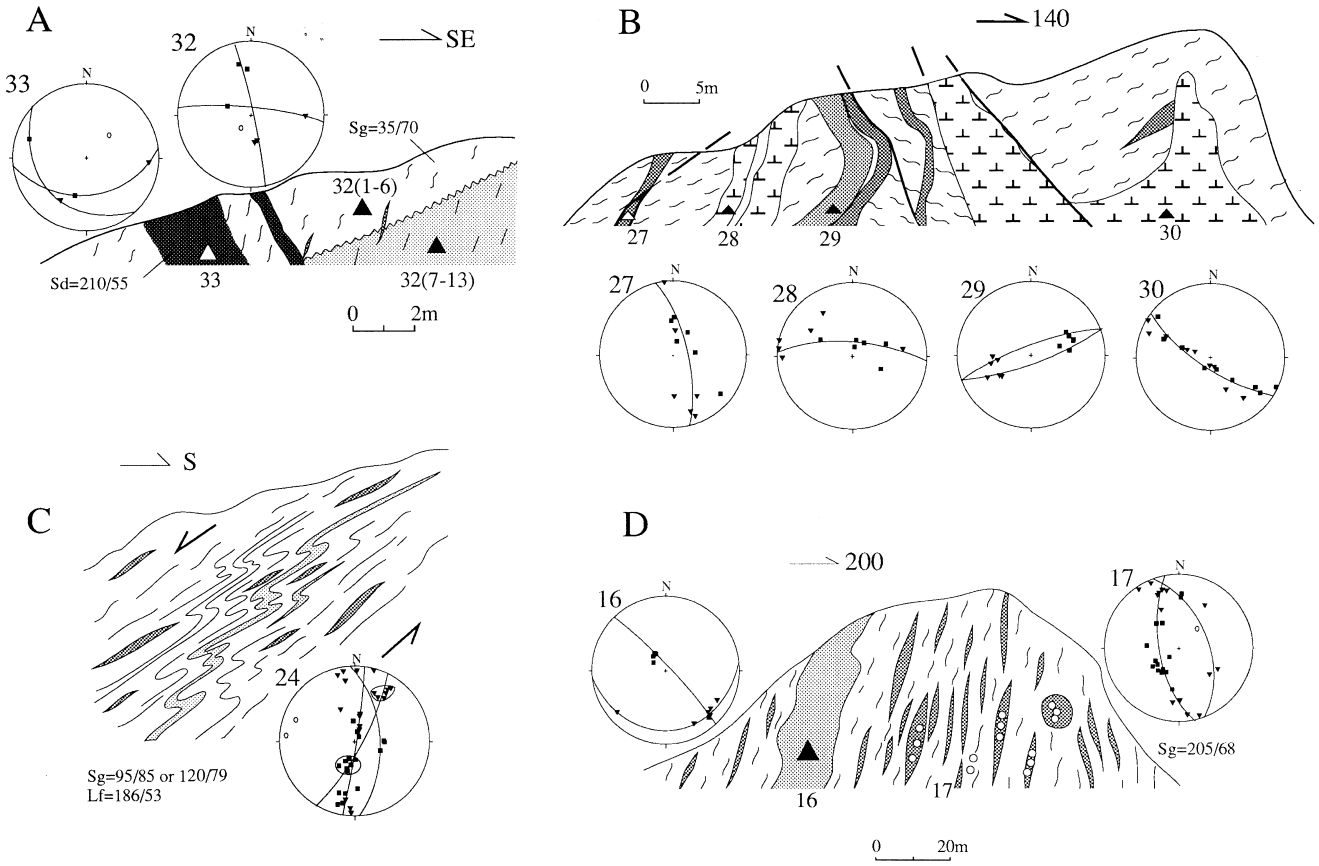


Fig. 7. Simplified illustrations of some sampled outcrops at which more than one site was located. All symbols are the same as for Fig. 2. Sg = gneiss foliation; Sd = dyke orientation, Lf = plunge of fold axis. See descriptions in the text for details. (A) Outcrop from north Zhunhua area embracing site 33 (basic dyke) and site 32/7-13 (mafic granulite below) and 32/1-6 (felsic granulite above). (B) Outcrop from north Zhunhua area showing a dyke set and sites 27–30. (C) Outcrop from north Qianxi showing the folding of gneiss and site 24. D) Outcrop from Caozhuang area of western Qian'an showing banded gneiss and site 17 (cores 1–12 from mafic granulite lenses, 13–14 from felsic matrix gneiss) and Site 16 (basic dyke).

Table 3. Thermomagnetic determinations (saturation magnetisation, J_s, versus temperature T) provide a diagnosis of magnetic mineral type and can recognise alteration during heating. All granulite samples illustrate type 2a behaviour (Mankinen et al., 1985) with discrete Curie points close to 580°C due to Ti-poor magnetite (Fig. 6a). The final saturation remanence, J_{sf}, is less than 10% lower than the initial saturation remanence, J_{si}); thus the magnetic phase exhibits only slight fall in J_s during the heating and cooling cycle probably due to some oxidation of magnetite to hematite. In contrast, the amphibolites all show type 2c behaviour with $J_{sf} > 10\%$ less than J_{si}; a residual tail is often present indicative of hematite and/or a significant contribution of paramagnetic minerals to J_s (Fig. 6a). The BIF have type 3 behaviour in which the cooling curve crosses the heating curve to give J_{sf} not more than 20% higher than J_{si}. This effect is more difficult to explain but is almost certainly related to behaviour of a titanomagnetite mineral; it is possibly due to a Ti-enriched phase not recognisable on the heating curve converting to magnetite which has higher J_s.

Study of magnetic susceptibility at low temperatures (KLT analysis) cools the rock to the temperature of liquid nitrogen (−196°C) and records susceptibility during subsequent return to room temperature. The susceptibility of multi-domain (MD) grains is controlled mainly by magnetocrystalline anisotropy (Uyeda et al., 1963) which is strongly temperature dependent: magnetite undergoes a structural change known as the Verwey transition at −150°C. At this point the magnetocrystalline anisotropy constant, k_1, is minimised; a peak in susceptibility near this temperature is diagnostic of MD magnetite and defines Group 3 behaviour of Senanayake and McElhinny (1981). Provided that grain elongation is >10%, single domain (SD) grain susceptibility is dominated by shape anisotropy which produces preferential magnetisation along the length of an elongate grain to minimise magnetostatic energy (Pullaiah et al., 1975). Since crystal shapes remain essentially constant with decreas-

ing temperature, shape anisotropy is not dependent on temperature and KLT variations are small. Superparamagnetic (SP) grains show a susceptibility falling steadily with temperature due to the increase in thermal energy. The SD–SP threshold (the "blocking volume") also decreases with temperature; below $\sim -150^{\circ}C$ relaxation times of the increasingly-smaller SP grains extend beyond the period of measurement, the threshold is then effectively exceeded and the susceptibility stabilises (Maher, 1988). Paramagnetic grains obey the Curie Law and susceptibility falls exponentially as the temperature rises.

Three parameters used to quantify KLT behaviour are RS (Relative Susceptibility) = K_{-196}/K_{30}, PS (Peak Susceptibility) = K_{peak}/K_{30} and ΔS = PS–RS (Radhakrishnamurty et al., 1977). They give a semi-quantitative indication of the relative SD and MD grain contents: $\Delta S \sim 0$ corresponds to the presence of SD grains only whereas $\Delta S \sim 0.9$ corresponds to a predominance of MD grains (Thomas, 1992). Group 3 behaviour is subdivided into 3a and 3b (Table 3) with the major difference between the two being that RS is always less than 0.5 for Group 3a and always greater than this for Group 3b. Radhakrishnamurty et al. (1977) describe Group 3a behaviour as indicative of almost pure MD magnetite with effectively no Ti substitution ($x < 0.15$); Group 3b behaviour has been considered to reflect the presence of only MD magnetite (Senanayake and McElhinny, 1981), although Radhakrishnamurty et al. (1977) consider it to be the signature of a mixed grain size assemblage with a MD-dominated assemblage contaminated by small amounts of SD grains.

Results of KLT analysis are summarised in Table 3 and two curve types are shown in Fig. 6. The granulites exhibit Type 3a behaviour defined by RS mostly <0.5 indicating a dominance of pure MD magnetite with no Ti substitution, and group 3b behaviour indicating a mixed grain size of magnetite but with MD grains still dominating over SD grains (Thomas, 1992). The banded ironstones also exhibit dominant MD behaviour (Table 3). The amphibolites either show type 2 curves (Senanayake and McElhinny, 1981) indicating that paramagnetic minerals dominate the KLT expression, or have hybrid type 2/3 curves with detectable MD magnetite.

High temperature susceptibility (KHT) analysis observes the variation in magnetic susceptibility when samples are heated to temperatures above the Curie point. Dominant MD behaviour is characterised by a broad Hopkinson peak yielding the H curve type of Radhakrishnamurty and Likhite (1970). Grains with SD behaviour at room temperature convert to SP behaviour at higher temperatures so that the susceptibility peaks before dropping gradually to the Curie point; this kind of curve is therefore the signature of dominant SD grain properties. The same effect is produced by a decrease in Ti content of the magnetite and the maximum susceptibility is attained at progressively higher temperatures as Ti is reduced (Dunlop, 1974; Appel and Soffel, 1985). Domain structure is sensitive to temperature changes and it is also common to find that irreversible changes take place such that a SD heating curve converts to an H type (MD) curve on cooling; this is referred to a quasi-single domain (QSD) type behaviour. The granulites and BIF facies exhibit H type behaviour (Fig. 6c) with small amounts of alteration during heating. KHT results thus reinforce the results of KLT analysis to show magnetic grains are dominated by MD grain assemblages. The response of the amphibolites to KHT analysis was too weak to quantify (Table 3).

Magnetic hysteresis was determined by subjecting samples to a cycled magnetic field within a vibrating sample magnetometer. The form of the hysteresis curve is expressed by the magnetisation at saturation (M_s), the residual magnetisation when the applied field is reduced to zero (M_{rs}), and the back field required to reduce this saturation to zero (this is the coercive force, H_c). If $M_{rs} \sim 0$ the result is a straight line of positive slope (i.e. there is no hysteresis), magnetisation is proportional to applied field and the behaviour is essentially paramagnetic. M_s and M_{rs} are dependent on the concentration and type of the magnetic minerals present; M_s is independent of grain size but M_{rs} is smaller for M_D than for SD grain properties. Hence the ratio of M_{rs}/M_s is a useful indicator of domain states: if $M_{rs}/M_s < 0.1$ the sample is dominated by MD grains, if $M_{rs}/M_s > 0.1$ small but significant fractions of SD grains are indicated within a dominant MD assemblage. Provided that these mixed domain sizes are present within magnetite only (indicated by M_{rs}/M_s values of 0.02–0.5), the equation:

$$X_{MD} = (0.5 - (M_{rs}/M_s))/0.48$$

yields an estimate of the fraction of MD grains present (Thomas, 1992) (Table 3).

The results of hysteresis analysis (Table 3) show that the granulites in the Huabei Shield have saturation magnetisations an order or more stronger than the amphibolite facies. A saturation achieved in low applied fields and the M_{rs}/M_s values indicate that magnetite is the mineral responsible for hysteresis in the granulites; the X_{MD} ratios identify a predominant MD character to these grains. In contrast, a barely perceptible hysteresis in samples from the amphibolite facies indicate that paramagnetic minerals dominate the magnetic response here.

Thus the rock magnetic tests show that pure magnetite grains with annealed properties stable to heating are responsible for the ferromagnetism of the granu-

lites. KLT, KHT and hysteresis studies all support the conclusion that the magnetite grain structure is predominantly or entirely MD. Thus the stable magnetic behaviour of these rocks (Section 5) indicates that either only a small fraction of the finer grain sizes contribute to the remanence in this facies, or (more probably) that domain walls in MD grains are pinned by effects such as grain shape. The significance of grain shape is implied by the reports of blade and needle-like grain forms in these facies elsewhere (e.g. Morgan and Smith, 1981; Piper, 1987; Zhang (J.S.) and Piper, 1994).

4. Palaeomagnetism and magnetic fabrics

The metamorphic basement outcrop from Miyun eastwards to the Bohai Sea (Fig. 1) is intruded by a number of granite plutons of Mesozoic age. The largest of these is the ~170 km^2 Yunmeng Shan Granite pluton (119–114 Ma, K–Ar) which has been sampled near the southern margin of the outcrop (Site 1, Fig. 2). With k_1/k_3 values of ~1.4, the granite has more prominent AMS than most of the metamorphic facies of the country rocks (Table 1). Also, unlike the majority of the metamorphic country rocks, fabric is

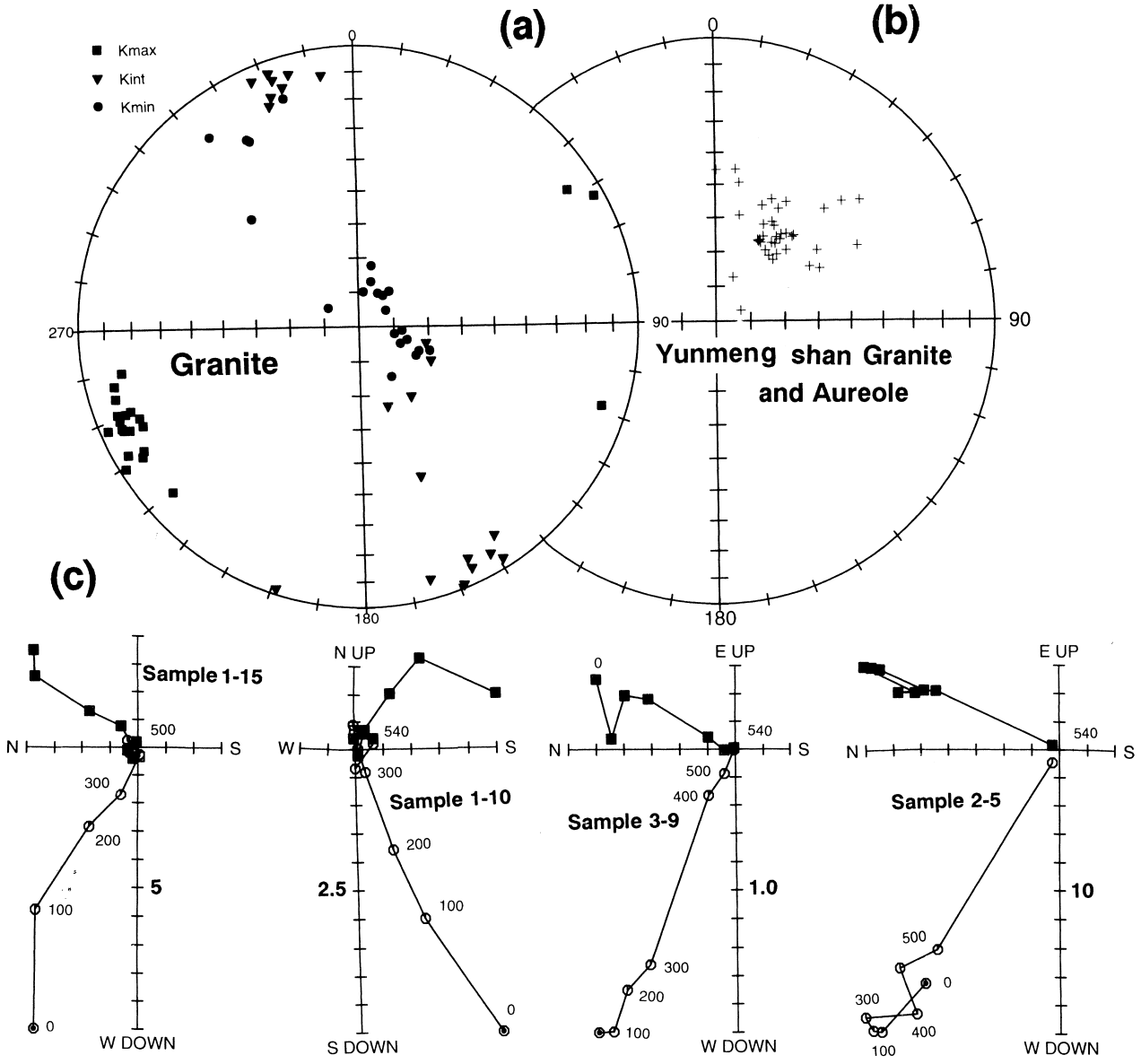


Fig. 8. (a) Direction of AMS axes (lower hemisphere) in samples from the Yunmeng Shan Granite. The symbols are as for Fig. 2. (b) ChRM directions in the granite and aureole at sites 1–3. (c) Orthogonal projections showing demagnetisation behaviour of samples from the granite and metamorphic rocks from the aureole. The closed squares are projections onto the horizontal plane and the open circles are projections onto the vertical plane. The figures are temperatures in °C and intensities are $\times 10^{-5}$ A m^2/kg.

Table 4a
Site and group mean palaeomagnetic results from metamorphic terranes of the North China Shield

Site no.	D°	I°	N	R	α_{95} (°)	k
(a) Miyun Terrane:						
1	33.4	57.5	15	14.52	7.2	29
2	37.3	49.4	7	6.74	12.8	23
3	36.6	62.4	19	18.85	3.0	123
4(i)	8.5	58.6	13	12.83	4.9	73
(ii)	37.8	59.5	3	2.93	12.4	99
5	65.2	61.1	7	6.88	8.6	51
6	17.2	27.5	11	10.83	6.0	58
7	0.0	58.2	10	9.71	8.6	33
(b) Qinglong (Amphibolite Facies) Terrane:						
8	6.2	46.2	4	3.79	25.0	15
10	336.8	20.6+	13	12.87	4.4	90
11(a)	357.0	26.9+	10	9.60	11.0	20
(b)	359.6	80.4	4	3.99	5.9	244
13	293.1	61.2+	6	5.87	11.1	37
15	335.0	8.5	3	2.94	21.7	33
(c) Zhunhua-Qianxi Terrane:						
16	355.4	−6.6	8	7.64	12.9	20
17	336.5	−9.3	4	3.73	29.1	11
18	211.5	67.9*	13	12.46	9.0	22
19	15.5	−5.1*	3	2.95	20.4	37
20(i)	38.5	−36.2	5	4.83	16.2	23
(ii)	131.7	68.8	7	6.85	9.6	41
21	63.5	−11.4	4	3.95	11.9	61
22	59.0	−48.8	11	10.84	5.8	63
23	55.5	−39.5	13	11.28	16.9	7
24	40.2	−47.5	20	19.19	6.9	23
25	312.1	−17.0	6	5.86	11.2	37
26	357.4	54.3	9	8.75	9.3	32
27	358.1	60.2	6	5.92	8.3	66
28	15.4	67.1	6	5.98	4.6	210
29	25.0	65.2	5	4.99	4.0	366
30	17.9	64.5	9	8.95	3.9	174
31	23.3	49.5*	11	10.74	7.5	38
32	27.5	57.2*	10	9.49	11.8	18
33	36.8	51.8*	6	5.97	5.0	180

Table 4b

Group mean calculations and pole positions: Pole Code:	D	I	N	R	α_{95}	k	Pole Position °E	°N	(dp/dm)
(i) Yunmeng Shan Granite and country rock aureole (Pole YG)									
YG	33.4	57.5	39	37.24	3.4	49	200.9	64.2	(3.6/4.9°)
(ii) Granulites, East Miyun Terrane (Poles MY1-3);									
MY1	51.9	64.7	13	12.69	6.7	39	171.4	48.5	(8.7/10.8°)
MY2	17.2	27.5	11	10.83	6.0	58	261.8	60.1	(3.6/6.6°)
MY3	1.5	57.5	20	19.60	5.6	51	128.7	84.5	(4.9/6.7°)
(iii) Amphibolites, Qinglong Terrane (Pole AM1 derived from sites 8, 10, 11, 13 and 14 in situ):									
AM1	16.0	31.0	20	19.56	5.0	43	250.6	68.5	(3.1/5.6°)
(iv) Granulites, Qian'an Region (Poles QN1 and QN2):									
QN1	356.4	−6.8	16	14.83	10.7	13	124.0	−46.4	(5.4/10.8°)
QN2	215.2	70.8	17	16.04	9.0	17	99.3	10.0	(13.6/15.6°)
(v) Granulites, Northeast Qianxi Terrane (Poles QX1-QX3):									
QX1	312.1	−17.0	6	5.86	11.2	37	172.1	−23.9	(6.0/11.6°)
QX2	44.2	−45.0	41	39.49	4.4	26	79.3	−11.3	(3.5/5.6°)
QX3	148.9	68.3	10	9.70	9.0	30	137.1	−6.1	(12.8/15.2°)
(vi) Granulites, North Zhunhua Terrane (Poles ZH4):									
ZH1	18.7	58.9	60	58.90	3.4	30	205.8	75.7	(3.8/5.1°)

D and I are the mean declination and inclination derived from N samples and R is the magnitude of the resultant vector. α_{95} is the radius of the cone of 95% confidence about the mean direction and k is the Fisher precision estimate (=(N-1)/(N-R)). Dp and dm are the radii of the oval of 95% confidence about the pole position in the colatitude direction and at right angles to it respectively. * Adjusted for tilt of nearby Jixian unconformity; + adjusted for local tilt of volcanic units

prolate and hence probably dominated by magnetite grain shapes.

Most samples have dual component remanence structure with a low blocking temperature (lbt) component of variable direction removed in the initial treatment (100°C) and a second more distributed component removed by 500°C. These ChRM's have NE + directions (e.g. samples 1–10 and 1–15 in Fig. 8c) with a few components veering towards a more easterly declination (Fig. 8b). All are of normal polarity; the mean declination is directed NE whilst the inclination is close to the present geomagnetic field inclination (+60°) at this latitude (Table 4a and 4b).

Sites 2 (dyke) and 3 (granulite) are in the metamorphosed country rock within 10 km of the granite margin and within its probable thermal influence. The dyke has AMS foliation subparallel to the dyke plane and horizontal k_1, whilst the magnetic foliation in the granulite is in the plane of the gneissosity (Fig. 2). Thermal overprinting by the granite is indicated by a dominance of NE + ChRM's (e.g. samples 2–5 and 3–9 in Fig. 8c).

The Yunmeng Shan pluton is located within the monoclinal limb at the southern margin of metamorphic basement outcrop in the Huabei Shield. It has been emplaced in the footwall of the Hefangkou Fault and exposed by core complex uplift during extensional tectonism (Davis et al., 1996). The sampled part lies below a (Shuiyu) mylonitic shear zone which defines the roof of the core complex and dips at ~20° south eastwards. The AMS fabric does not record this shear zone; instead the $k_1 - k_2$ AMS plane dips at a shallow angle towards the SW and is evident in the easterly dip of k_3 normals to this plane (Fig. 8a). Comparable fabric orientations are recorded in the granite along the border of the pluton (Davis et al. 1996). It is possible that this fabric is related to primary magmatic flow; on the assumption that the $k_1 - k_2$ plane was horizontal at the time of cooling of the granite body, the remanence direction adjusted for dip is D/I = 25.2/67.2° which yields a pole position at 170.6°E, 70.1°N plotting on the China APW path between 150 and 100 Ma (Fig. 14) and compatible with the 119–114 Ma radiometric age. Consideration of the remanence without tilt adjustment would make no essential difference to this interpretation; movements of the body during emplacement to the present level are more difficult to evaluate: whilst it is likely that the entire region has undergone some rotation about a near-horizontal axis during footwall uplift, any simple rotation of the gran-

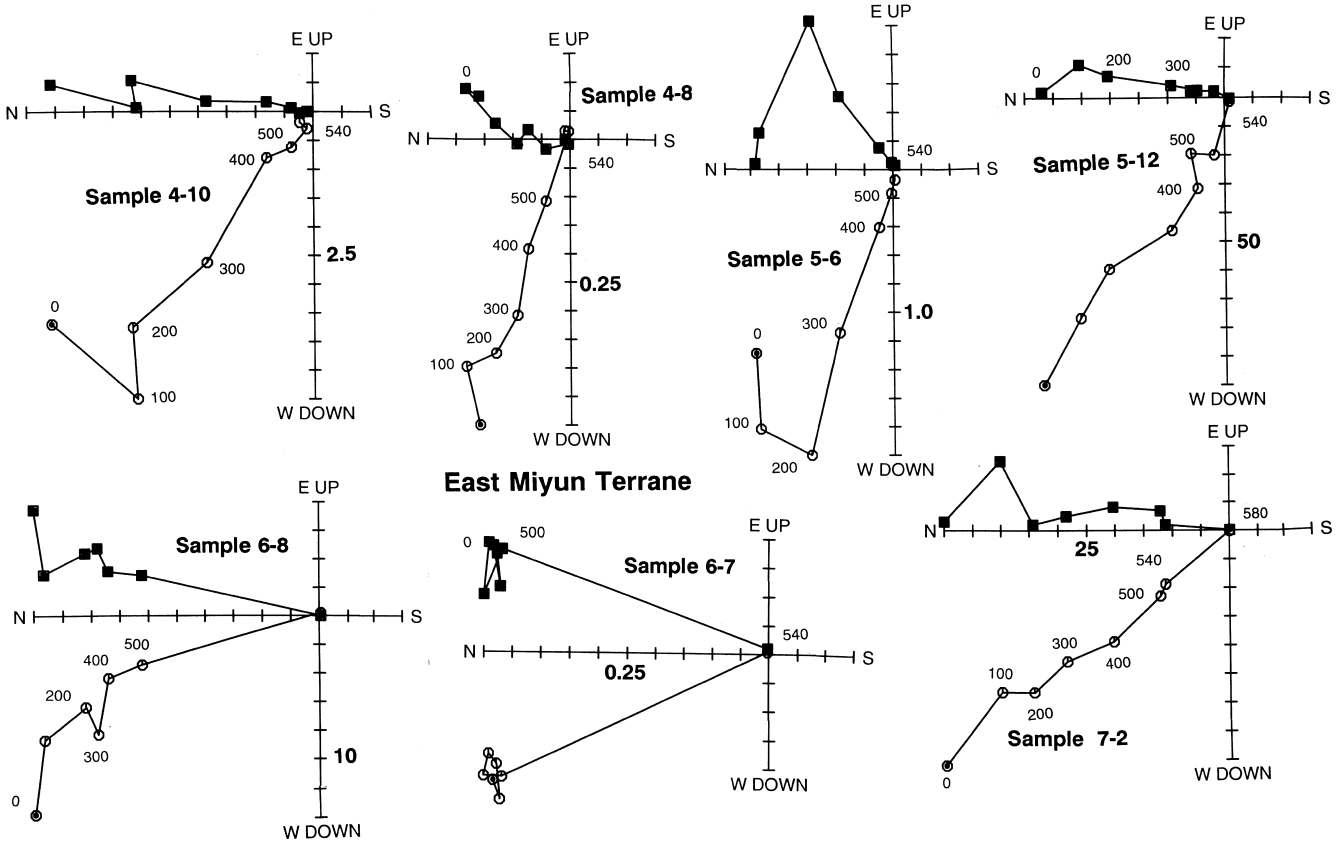


Fig. 9. Palaeomagnetic results for thermal demagnetisation from metamorphic rocks of the East Miyun region shown as orthogonal projections. The symbols are as for Fig. 8(c).

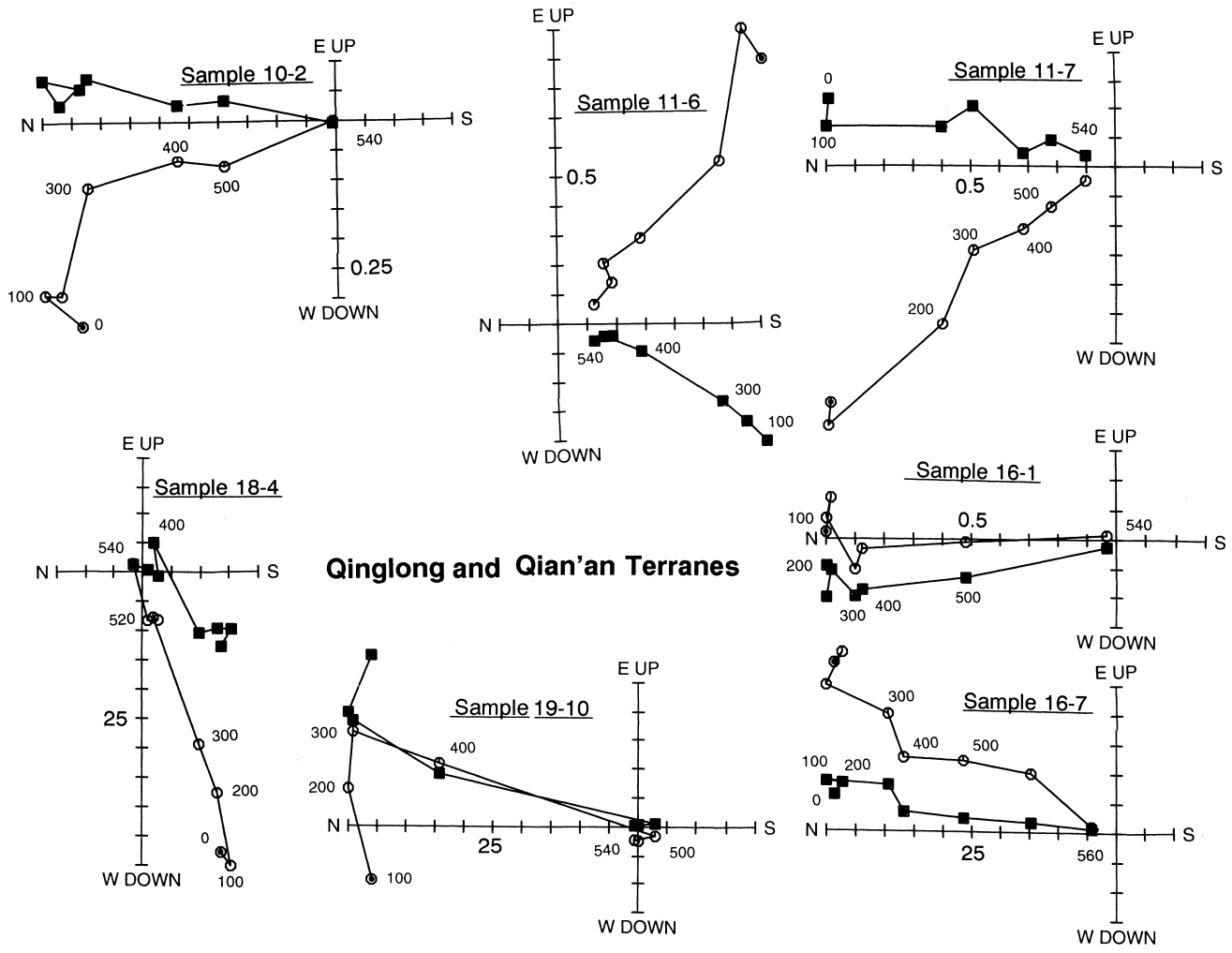


Fig. 10. Orthogonal projections illustrating palaeomagnetic results from metamorphic rocks of the Qinglong and Qian'an regions. The symbols are as for Fig. 8(c).

ite remanence about the 60°E strike of the Shuiyu shear zone increases the easterly declination of the remanence and shifts the pole position away from the Mesozoic–Recent APW path. Magnetisation of this body during the Cretaceous Normal Superchron (118–83 Ma) can explain why only normal polarities are observed in granite and aureole; this age of magnetisation is also compatible with the inferred Late Cretaceous uplift on the Hefangkou Fault (Davis et al., 1996).

Partial recovery of magnetisations unrelated to the granite is evident at greater distance from the contact: sites 4 and 5 include ChRM's with more northerly directions, sometimes following removal of a granite overprint by 300°C (Sample 4–8 in Fig. 9). AMS in granulites of this terrane is constrained to regional metamorphic fabric with the $k_1 - k_2$ plane in the foliation and a steep k_1 plunge in the observed lineation (Site 5). An exception is Site 4 which lies close to a late ductile shear zone retrogressed to amphibolite facies; the $k_1 - k_2$ plane at this location corresponds to the shear zone wall (Fig. 2).

Ironstones of Site 7 have discrete magnetisations which, although not subtracted until the Curie point of magnetite (Sample 7-2 in Fig. 9), are so close to the mean geomagnetic field (D/I = 0/ + 60°) that they cannot confidently be assigned an ancient origin (Table 3). The contact with a nearby rapakivi-type intrusion has similar declination but shallower inclination in narrow to distributed blocking temperature spectra (Samples 6-7 and 6-8 in Fig. 9), and has contrasting magnetic fabric (Fig. 2). A Mesoproterozoic volcanic suite spatially and temporally associated with rapakivi granite intrusions, including the sampled (Shachang) body, is linked to a mid-Proterozoic extensional tectonic regime (Yu et al., 1994). Age determinations on this intrusion are 1715 Ma (zircon U–Pb), 1588 Ma (biotite K–Ar) and 1311 Ma (Rb–Sr, Yu et al., 1994).

Sites 8 to 15 are in basic lavas, dykes and sheets (Table 1) from the Qinglong terrane in the eastern part

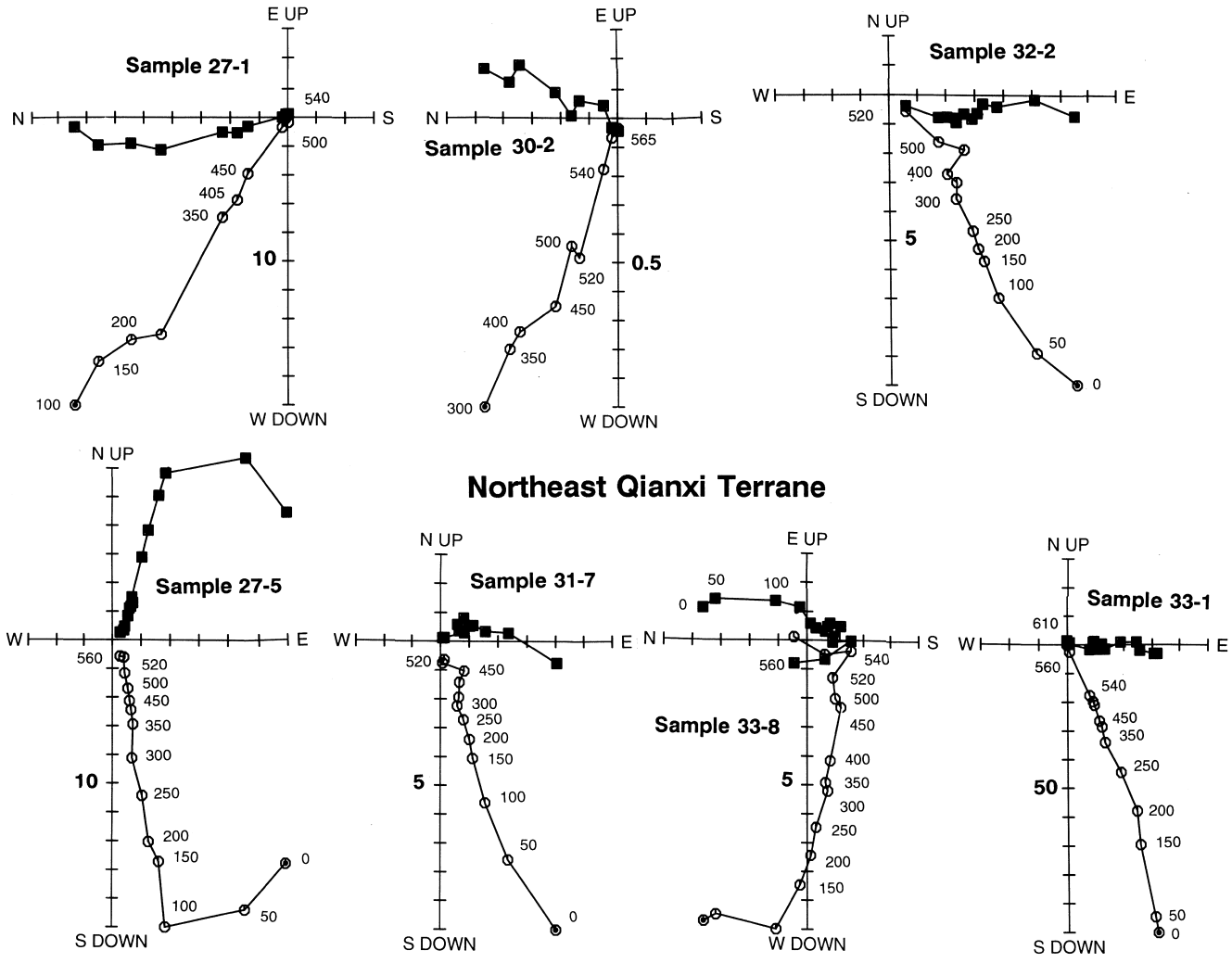


Fig. 11. Orthogonal projections illustrating palaeomagnetic results from the Northeast Qianxi region. The symbols are as for Fig. 8(c).

of the Zhunhua–Qianxi inlier (Fig. 1) and assigned to the Badaohe and Qinglonghe Groups of Archaean age; they comprise the highest crustal level investigated here. Metamorphic grades range from low to mid-amphibolite facies in the west to epidote to low amphibolite grade in the east and primary structures are widely preserved. The investigated section comprises a traverse from near the contact with a member of the Mesozoic granite suite outwards into country rock remote from younger magmatism (Fig. 3). Sites far from the granite tend to show Recent field components removed by 300°C to recover a N–NNE remanence of shallower positive inclination (Samples 10-2, 11-7 in Fig. 10); a few examples of opposite polarity occur (e.g. 11-6). Magnetisations at site 8 near to the granite contact are close to in situ remanence in the Yunmeng Shan Granite (Table 4, Fig. 8b and 14); elsewhere ChRM's have contrasting declinations and tend to have shallower inclination. Application of a palaeomagnetic fold test (McFadden, 1990) to the five stable sites (8, 10, 11, 13 and 154) shows that the hypothesis assuming a magnetisation acquired before deformation can be rejected at the 99% confidence level ($\S_{\text{tilt corrected}} = 4.377$, $\S_{\text{in situ}} = 3.573$ for $n = 5$). Hence magnetisations in the amphibolites postdate deformation and (with the exception of sites 10 and 13, Table 4) cannot be distinguished from the Mesozoic field.

AMS in the amphibolitised volcanics is dominated by foliatated fabrics with $k_1 - k_2$ planes dipping to the north west (Fig. 3). This is approximately orthogonal to the regional ESE dip of the succession and conforms to metamorphic foliation measured at outcrop. Close to the granite contact dip of the AMS foliation in dyke (Site 9) and host lavas (Site 8) changes to SE or is composite (Site 15, Fig. 3) and appears to be influenced by proximity to granite emplacement.

Sites 16–19 from granulite facies south west of Qian'an mostly possess ChRMs with northerly shallow direction subtracted by the Curie point of magnetite. These are often the only components present (Site 16

samples in Fig. 10) and typically have shallow negative inclination although some samples from site 19, are shallow positive. Comparable components are recorded in granulite and a cross cutting dyke, and directional data are collectively similar to the Datong–Huai'an granulite terrane to the west (Figs. 1 and 9) except that only one polarity is found at Qian'an (Fig. 13). An exception is the banded ironstone formation at Site 18 which, although showing comparable demagnetisation behaviour in a magnetite-held remanence, has a steep positive-directed ChRM contrasting with the granulites (Sample 18-4 in Fig. 10 and Table 4).

AMS fabrics within the Qian'an terrane are complex and appear to reflect deformation: at least two AMS foliation planes are present in the strongly deformed granulites at the Yugou iron mine (Site 19, Fig. 3). Site 16 retains a primary foliation near to the plane of the dyke whilst host granulites (Site 17) have foliation dipping steeply to the west in the basic lenses and to the east in the acid granulites (Fig. 7D). This is probably an example where the quartz-bearing felsic gneisses acted as a "softer" matrix and experienced stronger ductile deformation. The key observation is that the k_1 directions of these fabrics show no obvious relationship to the northerly shallow magnetic remanence that characterises these rocks; at Site 17 it is nearly orthogonal. Thus we suspect no strong influence of magnetic fabric on remanence here.

Samples of the basement terrane north-east of the town of Qianxi comprise basic and felsic granulites (represented by Sites 20–24 and 26) and a cross cutting basic dyke (Sites 25). The antiquity of the granulite protolith in this region is indicated by ~3500 Ma ages (Liu et al., 1990). The dominant feature of magnetic remanence is a NE-magnetisation (Samples 23-2 and 24-1 in Fig. 12) although declinations can range from northerly to easterly (Samples 23-4 and 22-2 in Fig. 12 and see Fig. 13). Some antiparallel lower blocking temperature components suggest a dipolar record and a dual polarity structure is present in some single cores (Samples 20-8 and 24-5 respectively in Fig. 12).

AMS fabrics in the North Qianxi terrane have variable orientations which mostly correspond to tectonic fabrics observed at outcrop. Differences are again recorded within individual sites at 20 and 22–24 between the AMS foliations in basic lenses and the felsic host (Fig. 3). Maximum susceptibility directions are unrelated to the dominant NE-magnetic remanence.

The remaining sites in the western part of this inlier occur north of Zhunhua (Sites 27–33, Fig. 3) and are in a sector invaded by numerous small granite plutons of Mesozoic age. Partial overprinted contributions of the Mesozoic to Recent field are present as lower blocking temperature components in acid and basic dykes at Sites 27–30 (eg. Samples 27-1 and 30-2 in Fig. 11) whilst possible ChRM's of more ancient origin have directions which are steeper (e.g Samples 27-5 and 30-2 in Fig. 11). At Sites 31–33 some ChRM declinations are more easterly (e.g. Samples 31-7, 32-2 and 33-1 in Fig. 11). The dyke at Site 33 has been metamorphosed with country rock of Site 32 and the dominant NE+ ChRM therefore records a common overprint. Sites 31–33 lie near the northern extremity of the basement inlier and dip northwards beneath the sub-Jixian unconformity. Although providing no significant test ($\S_{\text{tilt adjusted}} = 1.049$, $\S = 3.086$ for $N = 7$), adjustment of the steep ChRM's at these sites for tilt moves their ChRM's closer to the shallower ChRM's at the remaining Sites 27–30 (Table 4) and suggests that basement and cover may have been folded since acquisition.

Sites 27–30 from the same outcrop are deformed and sheared at the margins but preserve a post-tectonic AMS fabric contrasting with gently dipping metamorphic foliation in the country rocks at this location (Fig. 7B). Thus bodies 27–29 have steep foliation planes aligned close to the dyke planes whilst basic granulite lenses in gneisses of site 31 have weak magnetic fabrics correlating with metamorphic foliation (Fig. 3). Site 33 is an example of a composite fabric with one AMS foliation close to the dyke plane.

5. Regional magnetisations and pole positions

The magnetisation in medium-high grade basement terranes is always post-tectonic. The earliest opportunity for remanence acquisition occurs during post-orogenic collapse and cooling below the Curie point isotherm (~580°C for magnetite). Since the processes of remanence acquisition are very prolonged, even rock volumes as small as an individual core will have averaged short term variations of the magnetic field such as secular variation; this prolonged acquisition is enhanced by the lowering and broadening of the blocking temperature spectra consequent upon slow cooling (Pullaiah et al., 1975). Sample ChRM magnetisations are then integrated effects yielding palaeomagnetic poles; this is sometimes evident from recovery of a dual polarity structure during the demagnetisation of single cores. It can also mean that a varied remanence record is often recovered within a single palaeomagnetic site: local differences in magnetic mineralogy can result in remanence acquisition over time periods separated by many millions of years. Orthogonal plots from nearby cores shown in Figs. 9–12 illustrate this facet of basement magnetisation.

Accordingly no detailed analysis of data at site level is merited. Instead the populations of ChRM components form the basis of palaeomagnetic interpretation and are summarised as terrane by terrane distributions in Fig. 13. The distribution of sample

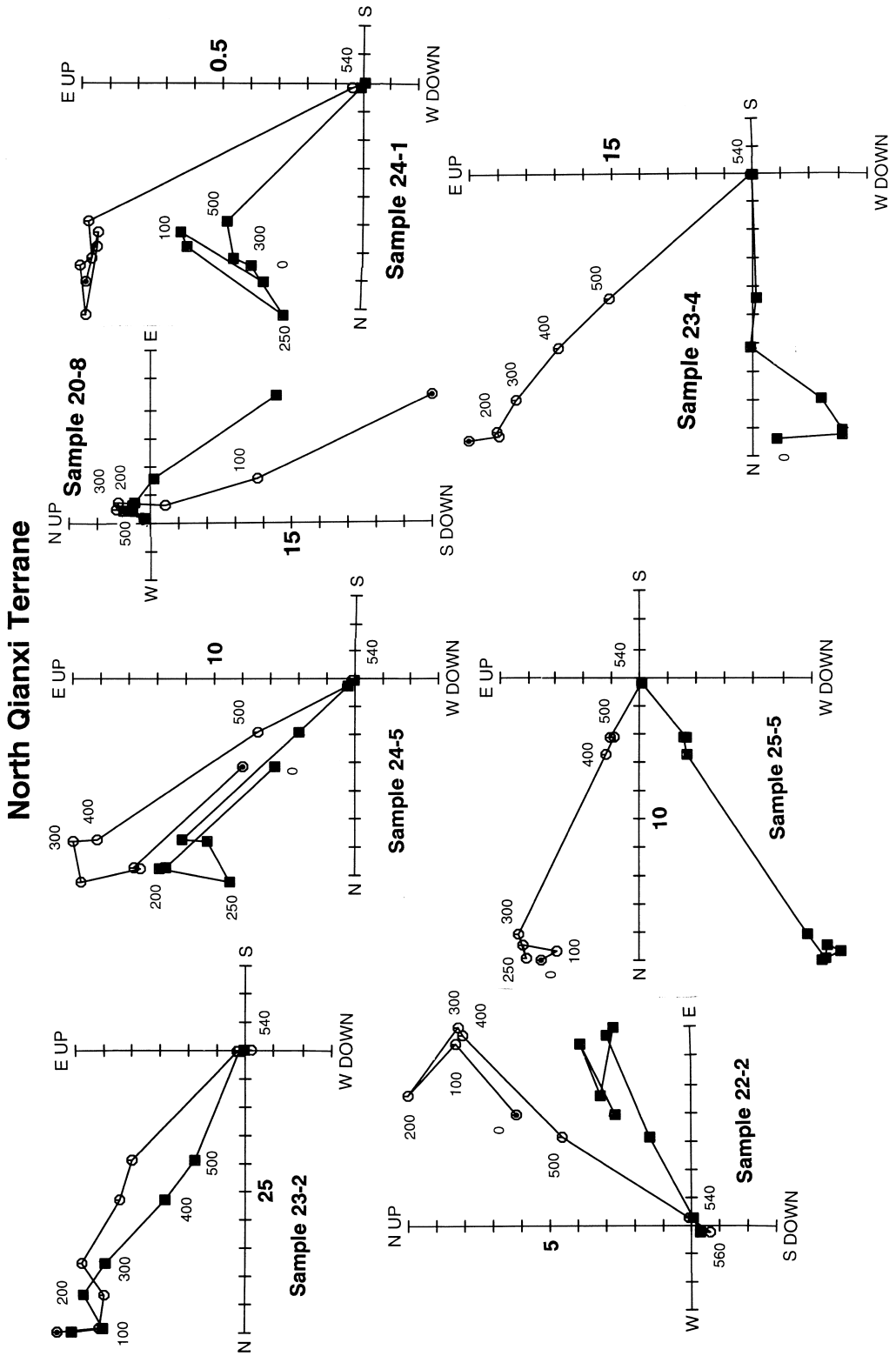


Fig. 12. Orthogonal projections illustrating palaeomagnetic results from metamorphic rocks of the North Zhunhua region. The symbols are as for Fig. 8(c).

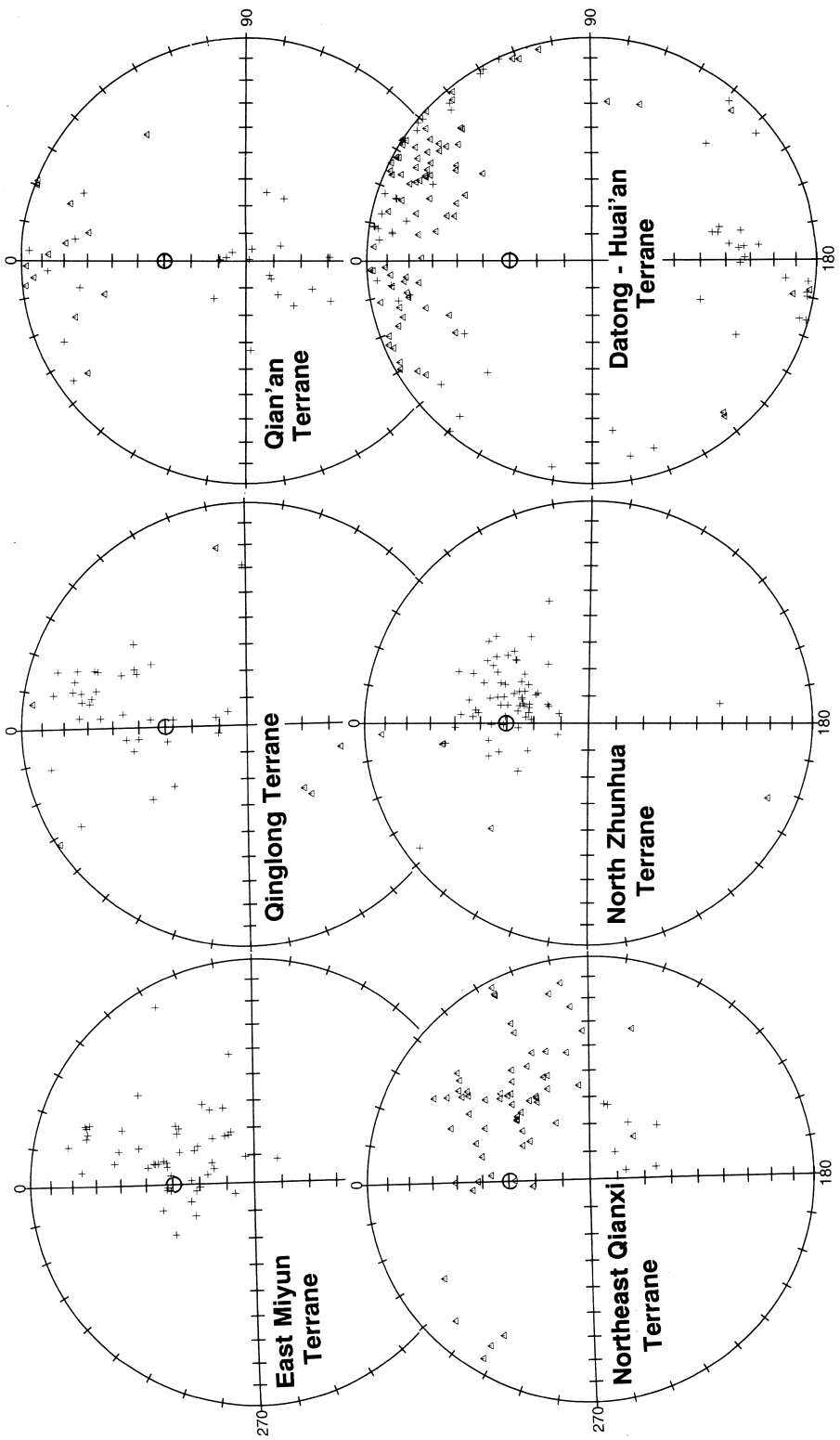


Fig. 13. Equal area projections showing sample ChRM directions from the metamorphic terranes of the North China Shield. Crosses are lower hemisphere directions and open triangles are upper hemisphere directions. The circled cross (positive inclination) is the mean direction of the present field in this region.

magnetisations in the Datong–Huan'ai terrane to the west of the study region (Zhang (Q.R.) and Piper, 1994) is also shown in Fig. 13 and is a typical record from a granulite basement. It identifies a long swathe of dual polarity magnetisations with northerly and southerly-directions of shallow inclination. The remaining record is observed to change along the Huabei Shield: although N–S shallow directions are sporadically developed, and may have recorded the same interval of uplift-related cooling as that identified in the Datong–Huai'an terrane, this is no longer the dominant feature of the distributions to the east (Fig. 13).

Since palaeomagnetic field tests are typically unavailable in basement terranes, interpretation of these magnetisations must have recourse to comparison with the established APW path of the North China Block. A comparison of data with the Phanerozoic path is

Table 5
Proterozoic palaeomagnetic poles from the North China Shield

Rock Unit/Formation/Terrane	Code	Age	Pole °E	°N	Si(Sa)	dp/dm Ref. or A_{95}	Ref.
Qianxi	QX3	2100–1900	317.1	6.1	(10)	13/15	1
Qian'an	QN2	2100–1900	279.3	−10.0	(17)	14/16	1
Qianxi	QX2	259.3	11.3	(41)	4/6	1	
Songting Fm.	STF	2060–2300 + +	270.3	−17.2	4(16)	—	4
Nansi Fm.	NS	2060–2300 + +	270.6	−20.6	6(26)	—	4
Nanzhangsi Fm.	NZF	2060–2300 + +	301.8	−35.0	6(24)	—	4
Datong–Huai'an	DHA1	2100–1900	230.6	20.1	6(48)	8/15	2
Granulites	DHA2	"	259.7	33.4	6(37)	6/11	2
	DHA3	"	286.2	48.1	8(57)	6/13	2
(Qian'an	QN1	2100–1900	304.0	46.4	(16)	5/11	1)
	DHA4	"	344.1	40.1	5(29)	11/21	2
(Qianxi	QX1	2100–1900	172.1	−23.9	(6)	6/12	1)
(Qinlong	AM1	2100–1900	250.6	68.5	(20)	3/6	1)
Miyun	MY2		261.8	60.1	(11)	4/7	1
Majiahe Fm.	MF		243.3	34.5	4(10)	12.9	5
Xushan Fm.	XF1	1829–1840 + +	221.3	44.0	5(9)	5.8	5
	XF2		238.0	44.2	4(9)	11.4	5
Dongjiao Group	DG	1810 + +	254.9	−25.0	7(28)	—	4
Changzhougou Fm.	CF	1837, 1848**	210.9	1.2	6(57)	16.0	5
	CHD1		250.5	−28.6	(14)	4.8	8
Chuanlinggou Fm.	CL1	1696 + + , 1785**	226.8	28.2	2(18)	9.1	6
	CL2	1717 + +	239.1	22.2	(39)	8.4	7
Tuanshanzhi Fm.	TF	1776**1606 +	249.4	7.2	(17)	6.1	6
Dahongyu Fm.	DR1	1625**	259.0	8.1	(11)	6.5	6
Dahongyu Fm.	DR2		283.1	13.9	3(31)	—	6
Gaoyuzhuang Fm.	GF1	1570–1678*	233.4	−0.5	3(25)	21.1	6
	GF2	1433**	246.7	0.7	(8)	8.9	6
	GF3	1425***	228.5	0.5	2(17)	—	6
Yangzhuang Fm.	YF1		222.5	16.4	4(62)	25.6	6
	YF2	1317***	215.7	21.2	2(20)	7.8	6
Wumishan Fm.	WR2		229.2	18.2	2(15)	10.6	6
NW Dykes	NWD	1229*1254 +	273.2	50.6	4(31)	4.5	3
N-S Dyke	NSD		238.8	43.7	2(17)	4.3	2
Tieling Fm.	TR1	1010–1234*	207.5	24.0	2(10)	9.4	6
	TG2		217.5	−9.4	(8)	19.1	6
Basic Intrusives	BI	1200*	210.9	−1.2	6(72)	16.0	6
Xiamaling Fm.	XB1	956*	209.7	−23.1	(9)	9.9	6
Jing'eryu Fm.	JGN1	810–926*	170.3	−12.5	(13)	13.4	6
	JR1-2	810–926*	142.7	3.8	(15)	4.7	6
Yongning Group	YG		185.5	−18.8	3(7)	14.7	6
Nanfen Fm.	NF	850*	121.0	−16.5	8(57)	9.5	4
Qiaotou Fm.	QF1		175.3	7.6	3(6)	15.8	6
	QF2		176.6	34.3	4(16)	9.1	7
Changlingzi Fm.	CF	732 +	156.5	46.6	9(74)	13.7	7

References are: 1, This study, 2, Zhang (J.S.) and Piper (1994); 3, Qian and Chen (1987); 5, Liu et al. (1984); 5, Zhang et al. (1993); 6, Zhang et al. (1991); 7, Zhang (1995a); 8, Zhang (1995b). Age determinations are: * K–Ar, ** Pb–Pb, + Rb–Sr, + + U–Pb, *** ^{39}Ar–^{40}Ar. Results in brackets are interpreted here as possible Phanerozoic overprints

made in Fig. 14 and with the post-metamorphic Proterozoic record in Fig. 15 and Table 5. Although post-uplift tilting of these rocks cannot be determined in the interior of the terranes, the Mesoproterozoic cover surrounding the metamorphic inliers has low dips seldom exceeding 10–15°. It is therefore unlikely that the basement rocks have been appreciably tilted since uplift to surface levels and in situ consideration of the palaeomagnetic directions and pole positions is associated with only minor uncertainties.

Magnetisations in granulites of the East Miyun Terrane have NE+ directions at Site 5 and part of Site 4; these are typically steeper than the Mesozoic–Recent field and yield pole MY1 removed from the Phanerozoic path (Fig. 14). The Shachang rapakivi intrusion has shallower inclination remanence (pole MY2) of possible Mesoproterozoic age. The remaining N+ magnetisations recognised in granulites at Sites 4 and 6, and ironstones of Site 7, have directions which cannot be distinguished from much younger field directions (Table 4); it is therefore unclear whether pole MY3 records a Mesozoic–Recent overprint or a primary magnetisation fortuitously near to the present field (Fig. 14).

With the exception of Sites 11(b) and 13 (Table 4), magnetisations in the Qinglong amphibolite terrane are northerly but mostly shallower than the Recent field. The pole AM1 is post-deformational and could also possibly be a Mesozoic overprint (Fig. 14). The Qian'an terrane has two ChRM populations comprising a steep positive magnetisation yielding pole QN2 and a shallow northerly magnetisation yielding pole QN1. The latter falls within the swathe of magnetisations observed in the Datong–Huai'an granulite terrane (Fig. 13) which is a basement outcrop devoid of Mesozoic magmatism and where Mesoproterozoic dyking is apparently the last thermal influence. Since it also contrasts with much of the Meso- and Neoproterozoic Jixian record, this swathe is probably a primary uplift and cooling-related acquisition at ~2100 Ma. This is also our preferred interpretation of pole QN1.

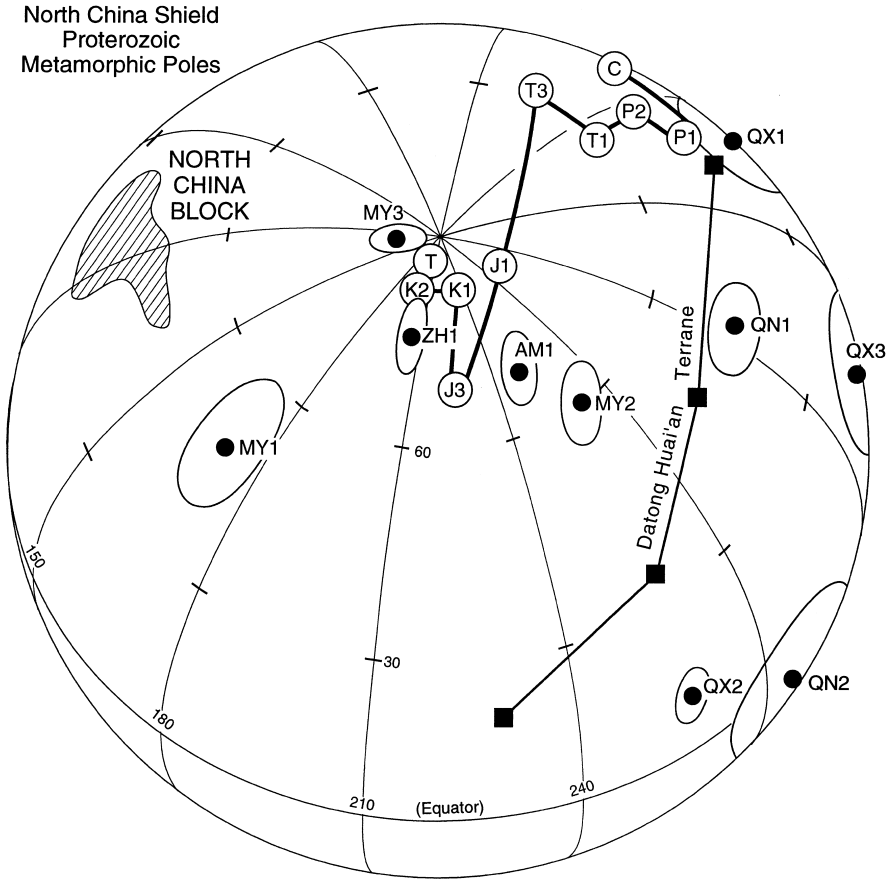


Fig. 14. Palaeomagnetic poles (black circles) from Precambrian metamorphic terranes of the North China Shield with ovals of 95% confidence. The poles are coded as in Table 4. Mean palaeomagnetic poles embracing the swathe of components resolved from the Datong–Huai'an terrane (Zhang (J.S.) and Piper, 1994) are shown as the tied black squares. The Phanerozoic apparent polar wander path from the North China Block as defined by mean poles of Zhao et al. (1996) is shown for comparison and links mean poles for Carboniferous (C), Early and Late Permian (P1, P2), Early and Late Triassic (T1, T3), Early and Late Jurassic (J1 and J3), Early and Late Cretaceous (K1, K2) and Tertiary (T) times.

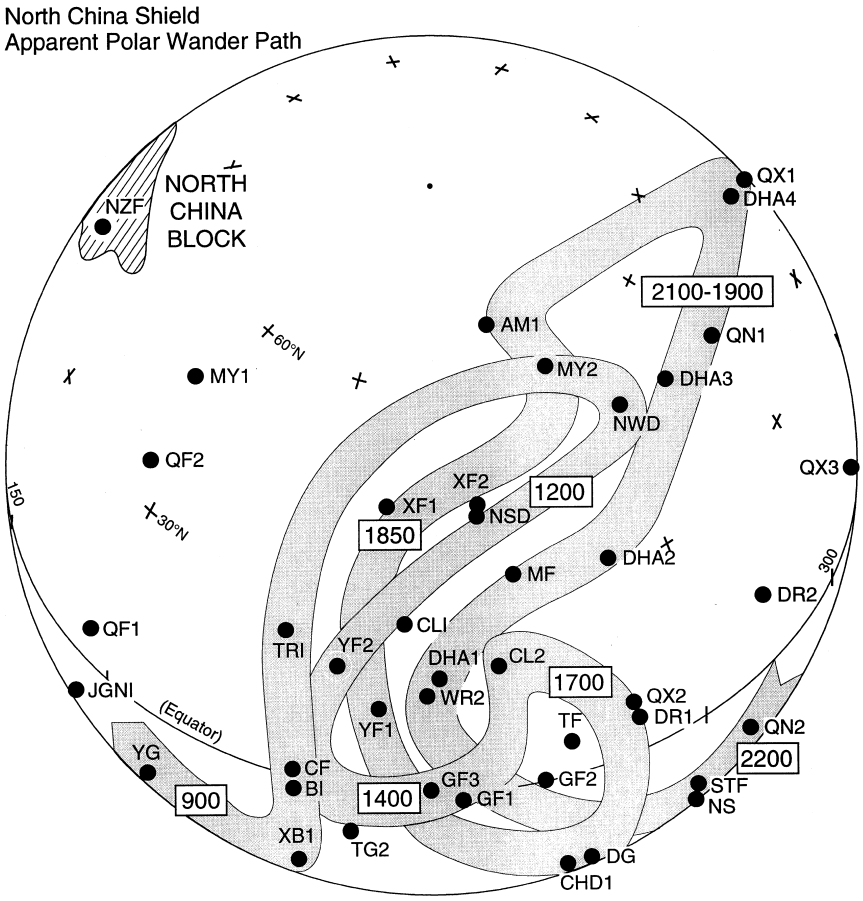


Fig. 15. Pole positions of inferred Proterozoic age from the North China Shield. An apparent polar wander path is proposed to embrace the time sequence of poles incorporating two loops (see Section 7). The poles are summarised and coded as in Table 5. Poles MY3 and ZH1 from this study are excluded as possible Mesozoic–Recent overprints (see text).

To the north in the granulite terrane of North East Qianxi characteristic magnetisations are predominantly NE. This record cannot be confused with the Phanerozoic field, and yields palaeopole QX2 plotting at the eastern extremity of Meso- and Neoproterozoic poles; hence we conclude that it is primary, and acquired during a stage of the uplift-related cooling. As is typical of this kind of imprint, it appears to record a swathe of declinations (Fig. 13), and probably a segment of APW, although the limited sample only merits a single group mean calculation. Minor populations of NW shallow components (yielding pole QX1) and steep positive components (yielding pole QX3) are also present. The former corresponds to the western extension of the swathe of magnetisation in the Datong–Huai'an terrane granulites (Zhang (J.S.) and Piper, 1994) and could be a primary magnetisation. However, it is also observed that this part of the swathe is close to the Upper Palaeozoic APW path (Fig. 14). No widespread Permian overprinting appears to be recorded in North China (Zhao et al., 1996) although Triassic overprinting (cf. mean poles T1 and T3 in Fig. 14) is widely identified further south in the vicinity of the Qinling Belt formed by welding of the North and South China Blocks (Kent et al., 1987; Wang (R.M.) and Van der Voo, 1993; Zhao et al., 1996, Zhang (Q.R.) and Piper, 1997).

The northerly positive magnetisation (mean D/I = 19/59°) is the exclusive feature of the North Zhunhua Terrane (Fig. 13). Although there is some indication that the northern margin of the basement has been folded since acquisition, this could have occurred during core complex uplift, and the pole ZH1 is indistinguishable from the Late Mesosoic field (Fig. 14). We therefore consider that it is probably a regional overprint acquired during the extensive Cretaceous plutonic igneous activity in this region (Fig. 3).

The fabrics recorded by AMS have their origins in mineral growth within a ductile regime under a regional stress field, as well as by subsequent microfracture infilling (Zhang (J.S.) and Piper, 1984). Magnetic remanence is typically a later feature acquired over a broad blocking temperature spectrum commencing at a temperature somewhat below the laboratory Curie points of the ferromagnetic carriers. Hence primary remanence acquisition by thermal or thermo-chemical

processes to impart a TRM or TCRM corresponds with the last part of the P-T-t loop in Fig. 4 and correlates with regional post-metamorphic uplift dated ~2100–1850 Ma in this region (Section 2). The magnetisations yielding pole QN1, QN2, QX1 and QX2 from the Qian'an and NE Qianxi terranes which do not correlate with known younger palaeomagnetic directions are considered to record palaeofields from this time period. The remaining record is indistinguishable from later field directions and cannot therefore be confidently assigned to the interval of uplift-related cooling. Indeed there is strong indication that much of this magnetisation record is a Late Mesozoic overprint acquired during emplacement of the North China Block into the eastern Asian collage and ensuing magmatism; this is most notably the case in the Miyun and North Zhynhua terranes (poles MY3 and ZH1) where poles are close to the Late Cretaceous APW path for North China and where granite emplacement of this age is extensive. We are unable to draw a specific link with the result from the Yunmeng Shan granite owing to uncertainties about its orientation during magnetisation (Section 3). Overprinting of early Mesozoic age is a possible explanation of magnetisations in the low grade metamorphic terrane at Qinglong (pole AM1).

6. Proterozoic apparent polar wander of the Huabei Shield

Inspection of Figs. 2, 3, 7 and 13 identifies no obvious bias of magnetic remanence by AMS lineation; also (and in contrast to the Datong–Huai'an terrane, see Zhang (J.S.) and Piper, 1994) we find no significant tendency for the swathes of component directions to lie within the planes of foliation. Hence we infer that fabrics in these metamorphic rocks have not deflected the remanence in a major way although some effect will no doubt be present. Dunlop and Özdemir (1997) summarise present evidence relating anisotropy to deflection of remanence; observational evidence is variable, but in theory, a susceptibility anisotropy of 22% (equivalent to the higher range of AMS values observed in this study) is equivalent to a maximum deflection of the remanence of 11.6°; this would influence only in detail the rather general conclusions that can be drawn from the remanence record in metamorphic terranes. The implication is that the pole positions derived from primary magnetisations in the metamorphic basement can be reliably used to define APW during the interval of uplift-related cooling.

The palaeomagnetic database from the Huabei Shield is summarised in Table 5. This compilation excludes earlier and poorly documented data but incorporates the studies based on modern methods of demagnetisation and component analysis (Zhang (H.M.), 1995a). The bulk of the Meso- and Neoproterozoic results are derived from formations of the Jixian Supergroup (Zhang (H.M.) et al., 1991; Zhang (H.M.), 1995b). The remaining data are from the metamorphic basement rocks (Liu et al., 1984; Zhang (J.S.) and Piper, 1994 and this study) and Meso-Proterozoic dykes (Qian and Chen 1987; Zhang (J.S.) and Piper, 1994).

Few of the studies summarised in Table 5 have associated palaeomagnetic field tests and most of the Jixian results are from single locations or small numbers of sites; assignment of ages to these magnetisations from sedimentary rocks is fraught with uncertainty. However, the bulk of the Jixian poles plot in the central Pacific region between 200 and 270°E and north of, or close to, the equator. The current basis for regarding this as a record of Meso-Neoproterozoic APW is a distinction between these data and the Carboniferous and later APW path, and specifically an absence of overprinting by the Late Mesozoic magmatic activity.

The bulk of the Proterozoic (~2200–900 Ma) poles from China plot as a distribution running across the north west Pacific and occupying a swathe which is approximately equatorial with respect to the North China Shield (Fig. 15). The limit of this distribution at poles QX1 and DHA4 overlaps with the Phanerozoic path and could therefore be influenced by later remagnetisation; the remainder of the swathe coincides with no known Phanerozoic data and can therefore be interpreted with some confidence as a Proterozoic record. These poles embrace a very long interval of geological time and, since their age control is broad, it may be premature to use them to define the shape of the APW path. However, there are several indications that at least two loops of comparable form are recorded here: firstly, poles from probable primary magnetisations in the metamorphic basement (NS, STF, QN2, QX2, NZF, and QX3) plot to the east of all the Jixian supracrustal data with the exception of two results from the Mesoproterozoic Dahongyu Formation (Fig. 15). Hence we recognise a general east to west APW motion between Palaeo- and Mesoproterozoic times. In the context of the uplift history of the North China Shield (Section 2), we suspect that the age of the older poles is somewhat younger that the lower limit of 2300 Ma given by Liu et al. (1984).

Secondly, the other key component of the Palaeoproterozoic record in the Huabei Shield is the Datong–Huai'an swathe (mean poles coded DHA in Fig. 15 and dated ~2100 Ma) which also includes pole QN1 and possibly pole QX1. The sense of motion along this swathe is not known for certain although Zhang (J.S.) and Piper (1994) suggest that it was from DHA1 → DHA4. These data are therefore shown in

Fig. 15 as a loop commencing with the other Palaeoproterozoic poles QN2, STF, NS and QX2 and returning via probable Mesoproterozoic poles from the oldest of the supracrustal rocks with assigned ages of ~1850 Ma (XF1, XF2, CF, DG, CHD1); poles MY2 and AM1 could fall on the return segment of this track or (as noted above) record Mesozoic overprinting. It also remains possible that the Palaeoproterozoic segment of this loop has the opposite trend commencing with the Datong–Huai'an swathe and moving towards the poles QX2, QN2, STF and NS.

Palaeopoles from the oldest of the Mesoproterozoic cover rocks (MF, XF1, XF2, CF, DG and CHD1) are dispersed and suggest the possibility of appreciable APW motion during these times. It is possible that subsequent APW was rather low however, because younger Mesoproterozoic formations of the Jixian Supergroup (~1700–1300 Ma) are distributed along an east to west trend of little more than 50° (poles TF, DR1, GF1, GF2, GF3, YF1, YF2 and WR2). Although it is not yet defined with clarity, a loop is then required to embrace poles from late Mesoproterozoic dyke swarms emplaced into the cold shield (poles BI, NSD and NWD in Fig. 15). Poles from Neoproterozoic formations of the Jixian Supergroup and elsewhere plot further to the west of the Meso- and Palaeoproterozoic poles and reflect a trend also seen in APW paths from other shields such as Laurentia and Baltica (Piper, 1987).

Thus the Precambrian palaeomagnetic record from the North China Shield appears to commence at about 2200 Ma and define a general east to west trend through more than 1200 Ma of Proterozoic times. We propose that at least two APW loops are incorporated within this general trend, although more well-dated magnetisations are required to define these (and possible other loops) more clearly.

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